Ophthalmology Workbook

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MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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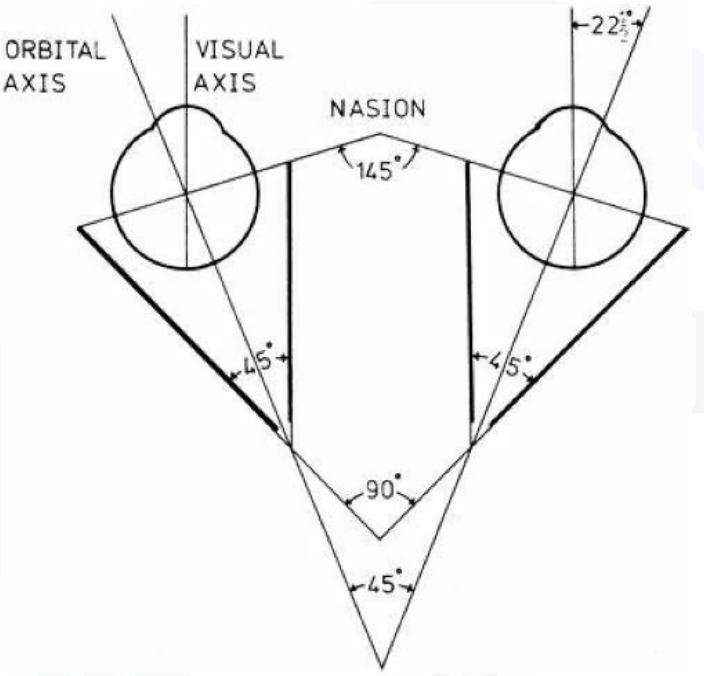
THE ORBITAL PYRAMID

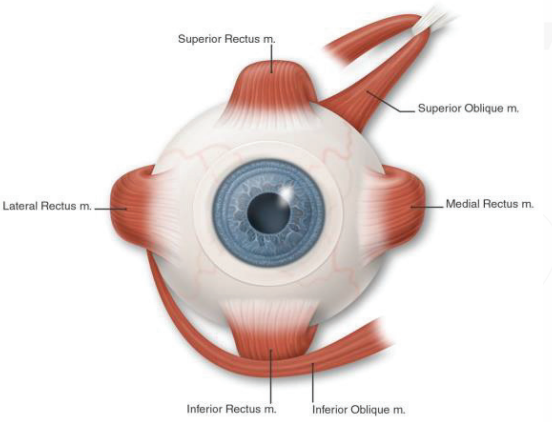
- Volume of each orbit -30 ml , pyramidal in shape
- Volume of each eyeball -6 ml

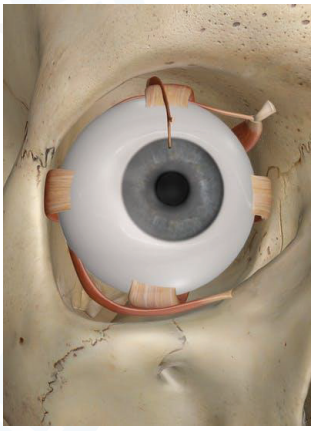


Active space

ORBIT

- Rule of 7 – 7 bones, 7 muscles, 7 nerves in orbit, 7 attachments of eyeball
- 6 EOM of eyeball, 1 of eyelid
- 6 movements-elevation/depression/abduction/adduction/intorsion/extorsion





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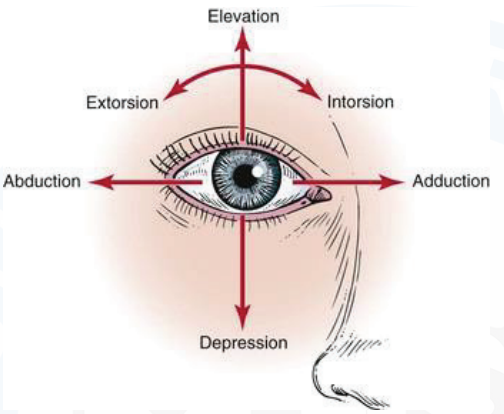
EXTRAOCULAR MUSCLES EOM

- Origin – annulus of Zinn (apex of orbit), except IO
- Actions - SINRAD
- Superiors (SR/SO) INtort, Rectus (SR/IR) ADDuct



EYELIDS

- Protect eyeball and spreads tear film
- Eyelid elevators – LPS, Muller's, -
- Only one muscle to close eyelid -

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NORMAL EYELID POSITION

- Normal upper eyelid covers 2mm of cornea
- Lower lid on -

PTOSIS

- Drooping of upper eyelid >2 mm

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CONGENITAL PTOSIS

- From birth till one year of age
- Simple ptosis
- Marcus Gunn/Jaw Winking ptosis

MARCUS GUNN / JAW WINKING PTOSIS

- Congenital ptosis with abnormal movements of upper lid with movements of jaw
- Due to abnormal connections between 3 CN and 5 CN

ACQUIRED

- Ptosis after one year of age
- Aponeurotic / Involutional – senile
- Neurogenic-Horner's, 3rd CN palsy
- Myogenic- Myasthenia , -
- Mechanical – Chalazion

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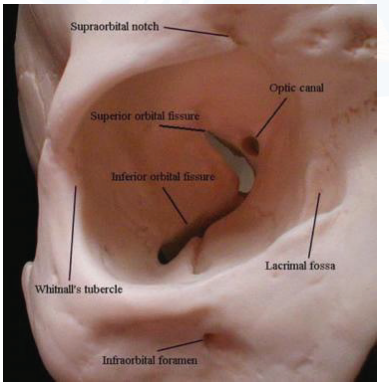
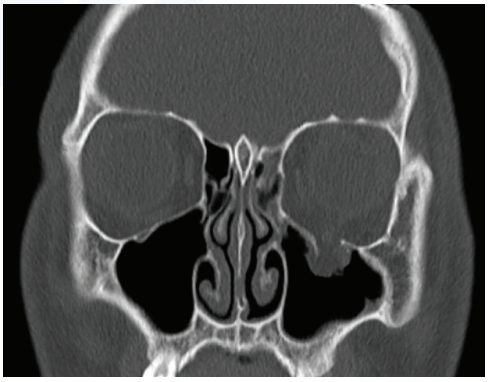
MANAGEMENT OF PTOSIS

- Good LPS action
- LPS resection

ORBITAL BLOWOUT FRACTURE

- Fracture of orbit without including rim
- Size of impact > size of orbital aperture
- M/c wall to fracture -
- Diplopia , Infraorbital anaesthesia, enophthalmos (DIE)
- Tear drop sign





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PROPTOSIS / EXOPHTHALMOS

- Causes – Thyroid eye disease , Orbital cellulitis
- M/c/c – Thyroid eye disease / Graves' disease

THYROID EYE DISEASE/GRAVES' D

- Females>males -40 to 60 years
- M/c presentation –
- Proptosis
- Enlargement of muscle belly with sparing of tendons - bottle sign
- Restriction of EOM – IMSL
- First muscle --

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ENOPHTHALMOS –RETRACTION OF EYEBALL

- Blow out fracture

ORBITAL TUMOURS

- Most common primary orbital tumour of children
- Dermoid cyst
- Most common secondary tumours of orbit in children
- Metastases from
- Neuroblastomas --

  Ewing's Sarcoma
- Most common primary tumour of the orbit in adults
- Cavernous haemangioma
- Most common secondary tumours of the adult orbit
- Metastasis from
- .....
- Lungs
- Prostate

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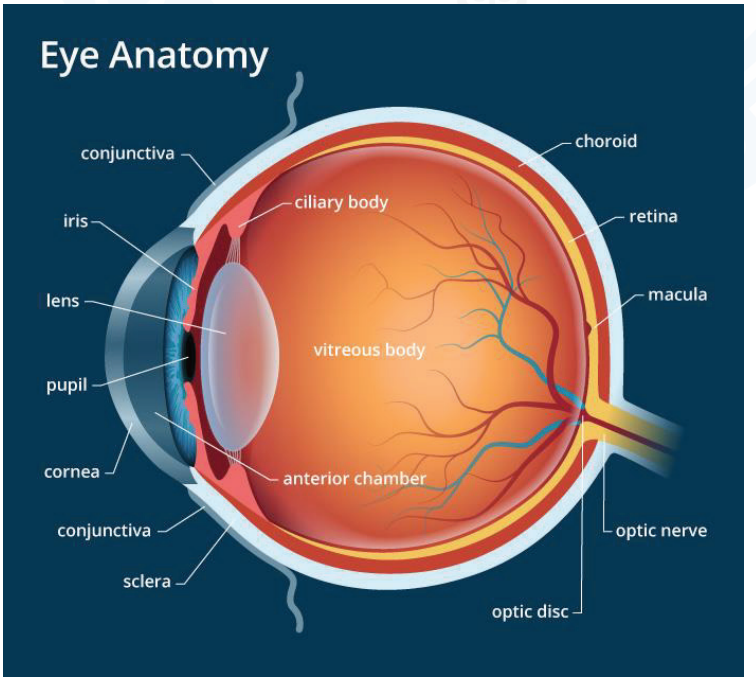
ENUCLEATION VS EVISCERATION

- Enucleation - Removal of eye leaving extraocular muscles and other contents of orbit intact

- Evisceration - Removal of contents of eye leaving sclera
- Removal of - and inner layers

HUMAN EYE – 3 CONCENTRIC LAYERS



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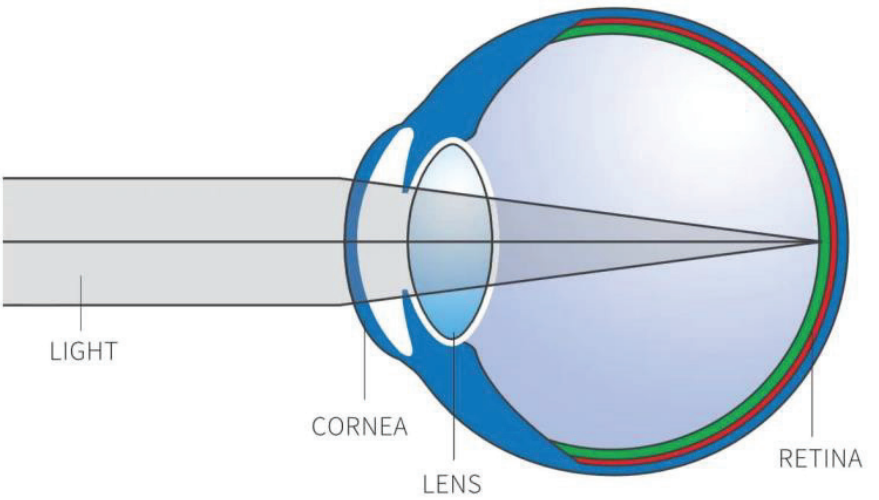
CORNEA/SCLERA

- Eyewall –Cornea/sclera
- Function – Protection of intraocular contents
- Made up of collagen , glycosaminoglycans GAG
- Cornea contributes 70% of power -43 D /60 Dioptres
- Lens 30 % --



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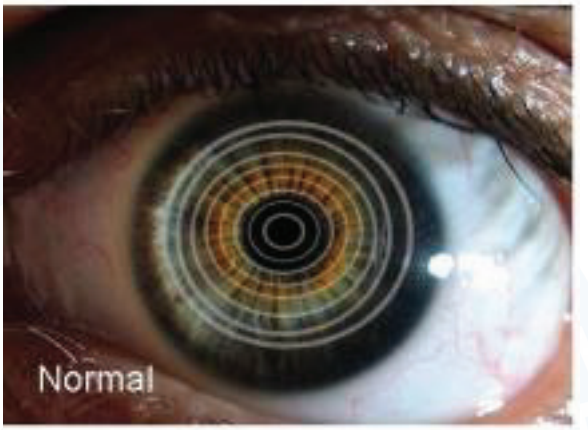
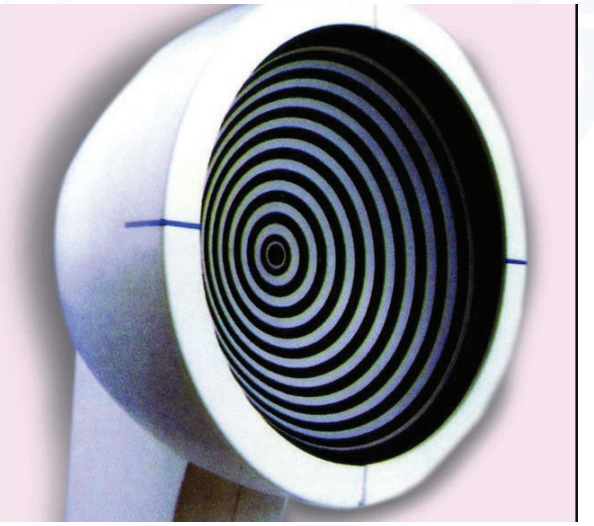
KERATOMETRY

- Measures corneal curvature – Placido's disc
- Detects Keratoconus





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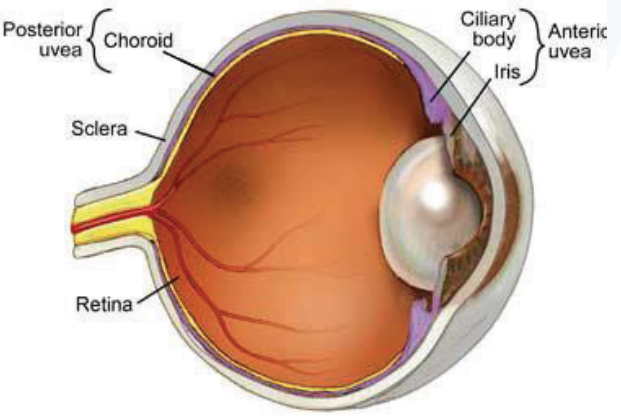
BENDING OF LIGHT

- Amount of light bent by a surface depends on
- Curvature of surface
- Refractive index of surface
- -
- Highest RI of eye –centre/nucleus of lens

UVEA(IRIS / CILARY BODY /CHOROID)



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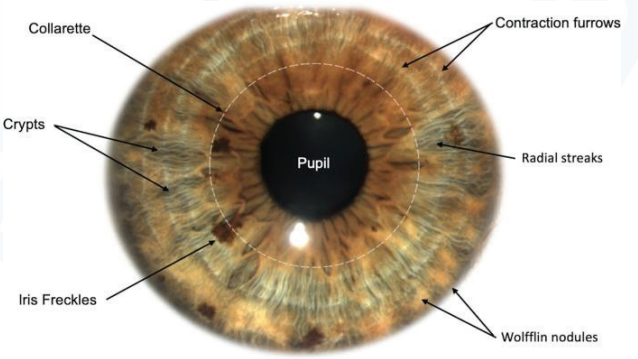
IRIS

- Named after Greek goddess
- Pupil contracts / dilates to control entry of light
- Sphincter pupillae /Dilator pupillae

IRIS STRUCTURES



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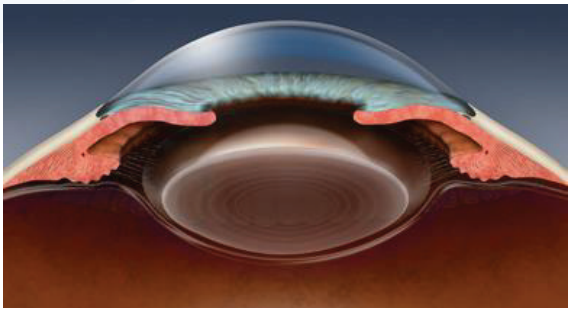
IRIS

- Iris pattern different for everyone normal pupil diameter 3-4 mm may vary from 1 mm to 9 mm

CILIARY BODY

- Anterior part folded –pars plicata
- Posterior part not folded – pars plana
  - Ciliary muscle - Accommodation
  - Aqueous humour production



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IRIDODIALYSIS –D SHAPED PUPIL

CHOROID

- From chorion of placenta
- Intraocular structure with the highest blood flow
- Thermoregulation/Nutrition to retina
- Drained by vortex veins

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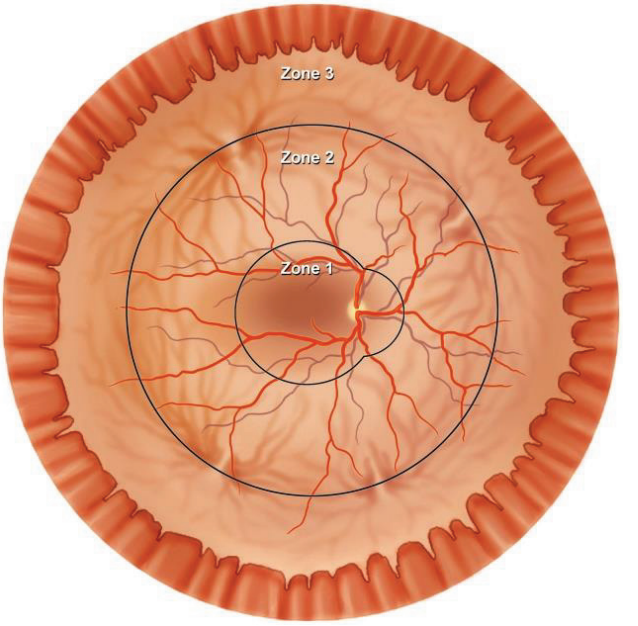
RETINA

- Inner most layer of eye – centre / periphery
- Periphery –ora serrata –thinnest part of retina
- Centre -3 structures – optic disc, fovea and macula
- Macula- centre of retina – from yellow colour
- Fovea – centre of macula
- Most sensitive to light/ Clearest , sharpest image/ 80 % - 90% light focusses on fovea
- Optic disc –beginning of optic nerve (ONH) ,only part of ON visible
- Optic disc diameter ODD = 1.5 mm = fovea



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MICROSCOPIC STRUCTURE OF RETINA

- Neurons and neuroglia
- 5 types of neurons , 3 types of neuroglia
- Rods and cones –photoreceptors –First order neurons
- Bipolar cells- - Order neurons
- Ganglion cells – - Order neurons
- Axons of RGC collected as optic nerve
- Horizontal cells
- Amacrine cells

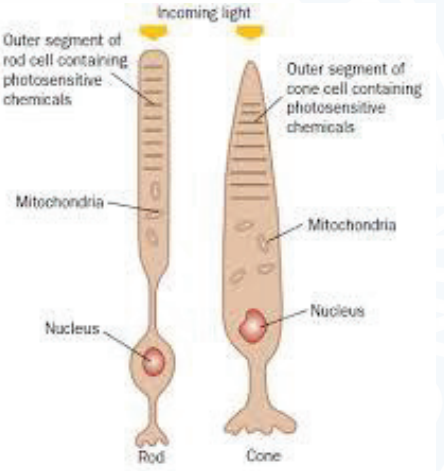

Figure 8.4.2 Basic structure of rods and cones

Rods	Cones
• Location Periphery	• Central foveal cones
• Number 120 million	• 6 million
• Only one type	• 3 types –Red green blue
• Peripheral vision	• Central vision
• ..... - vision	• ..... - vision
• Night vision	• Daylight vision

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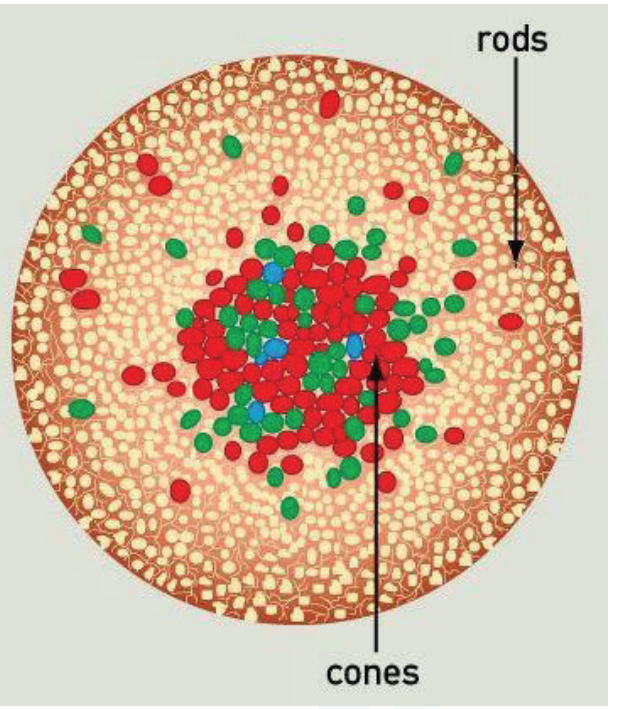
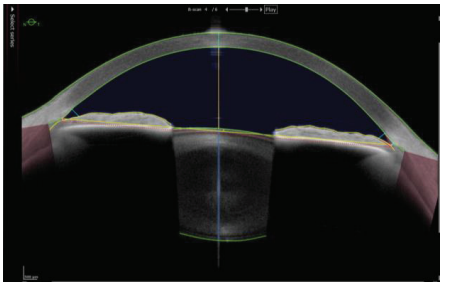
RODS AND CONES



ANTERIOR CHAMBER

- Space between cornea and iris
- Depth – average 3 mm
- Shallow –WEC
- Deep –young males
- Contains aqueous humour – nutrition to cornea , lens



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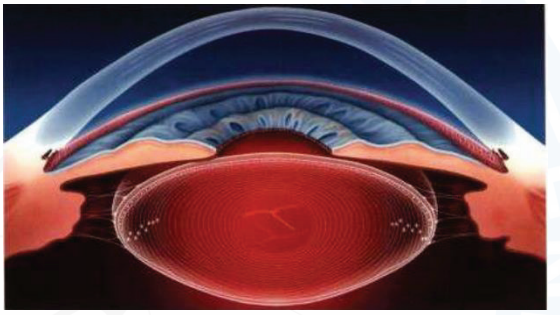
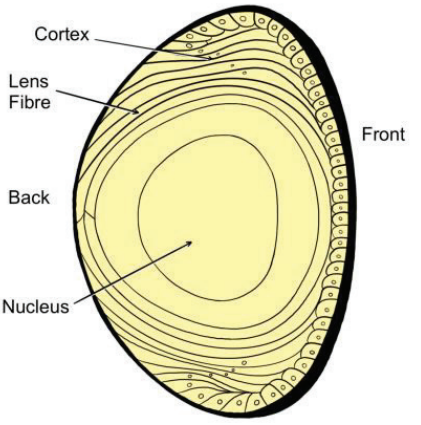
CRYSTALLINE LENS

- Biconvex , avascular, transparent structure located behind the iris
- Nucleus –centre, cortex – periphery , epithelial cells , lens fibres
- Functions – transmits and focuses light on retina , filters UV light , accommodation
- Held by zonules to the ciliary body





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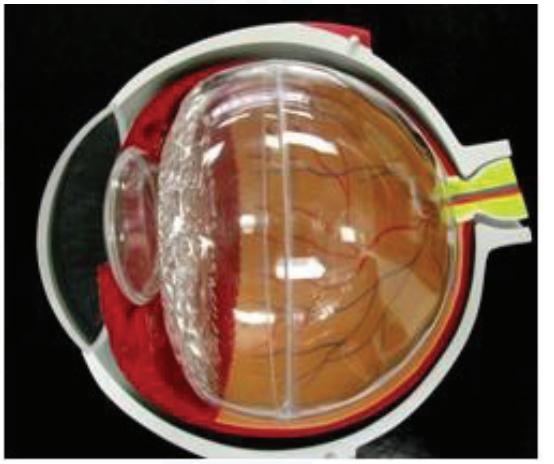
VITREOUS HUMOUR

- Clear jelly -like substance behind lens/front of retina
- Occupies 4 ml /6 ml – largest structure in eye
- Attached to optic disc margin , fovea , blood vessels of retina , vitreous base
- Strongest attachment of vitreous is to vitreous base – separation pathognomonic of ocular -



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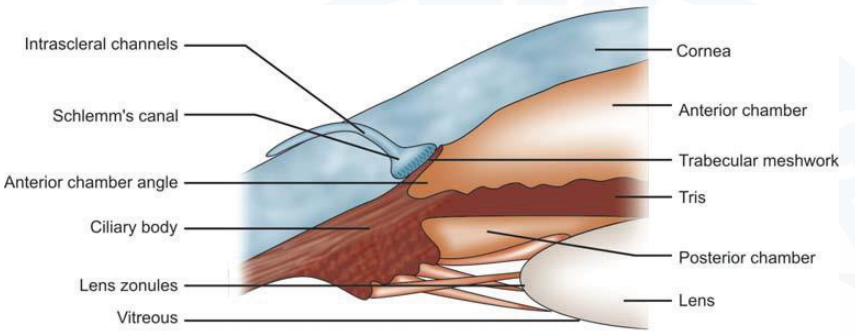
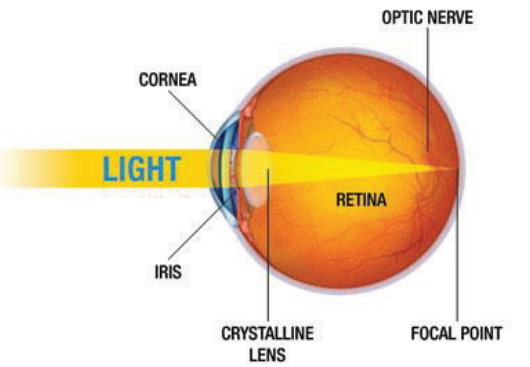
AQUEOUS CIRCULATION

- Ciliary body (pars plicata) – PC – pupil –AC – angle –trabecular meshwork – Schlemm's canal –episcleral veins
- Commonest site of blockage -



OPTICS OF THE EYE

- Light bends at 4 different surfaces as it enters the eye , 4 changes of media
- Air –Anterior surface of cornea(tear film)- Max refraction
- Cornea to aqueous
- Aqueous to lens
- Lens to -



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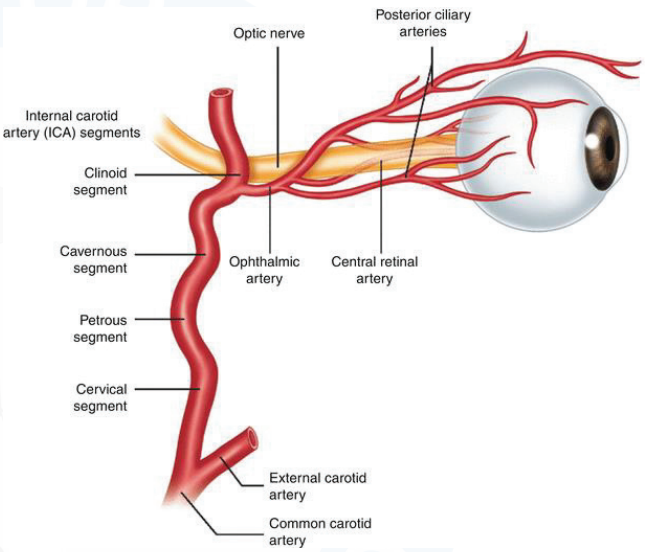
VASCULATURE OF THE EYE

- Principal artery of eye Ophthalmic Artery – 10 br
- Most critical – Central Retinal artery -CRA
- Principal veins – Central Retinal Vein / Vortex veins



NERVE SUPPLY OF EYE

- 6 Cranial nerves – II, III, IV, V ,VI,VII
- Autonomic nervous system -
- Parasympathetic – III nerve – Miosis, accommodation,
- Sympathetic - V nerve(trigeminal hitchhiking)

  – Mydriasis, Muller's muscle, Inferior tarsal muscle , --

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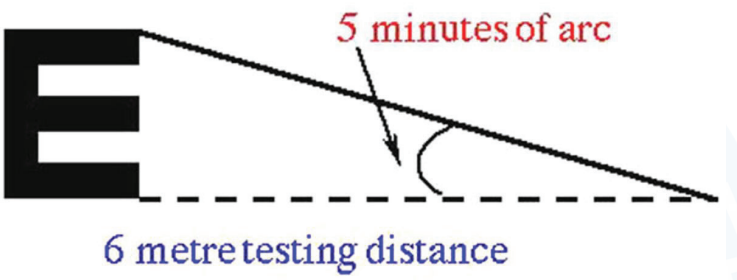
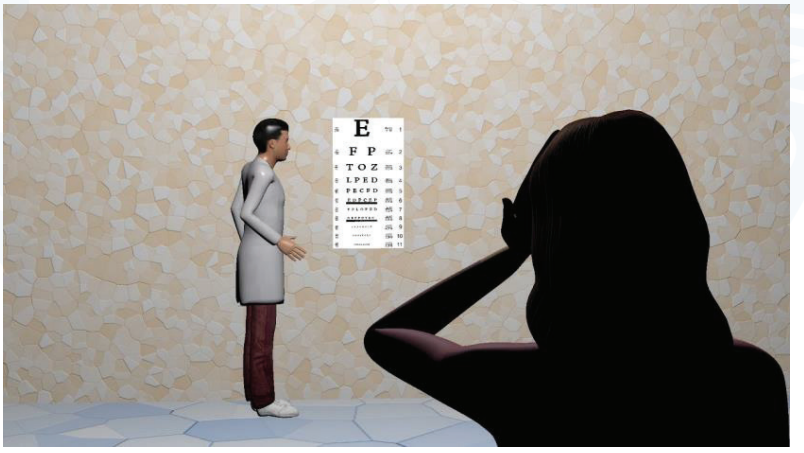
VISION/VISUAL ACUITY

- Standard normal vision – 6/6 or 20/20
- Distance between patient and chart / Distance between NORMAL person and chart
- Each letter subtends an angle of 5 minutes at nodal point of eye



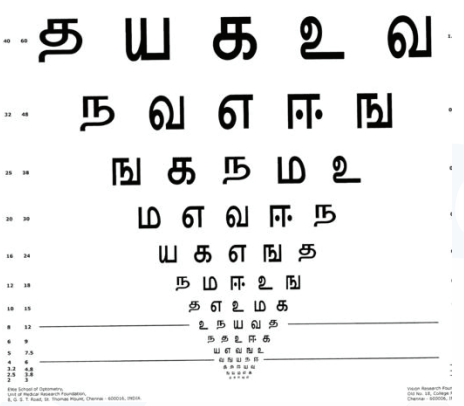


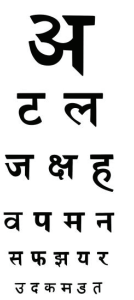
Active space

SNELLEN'S CHART

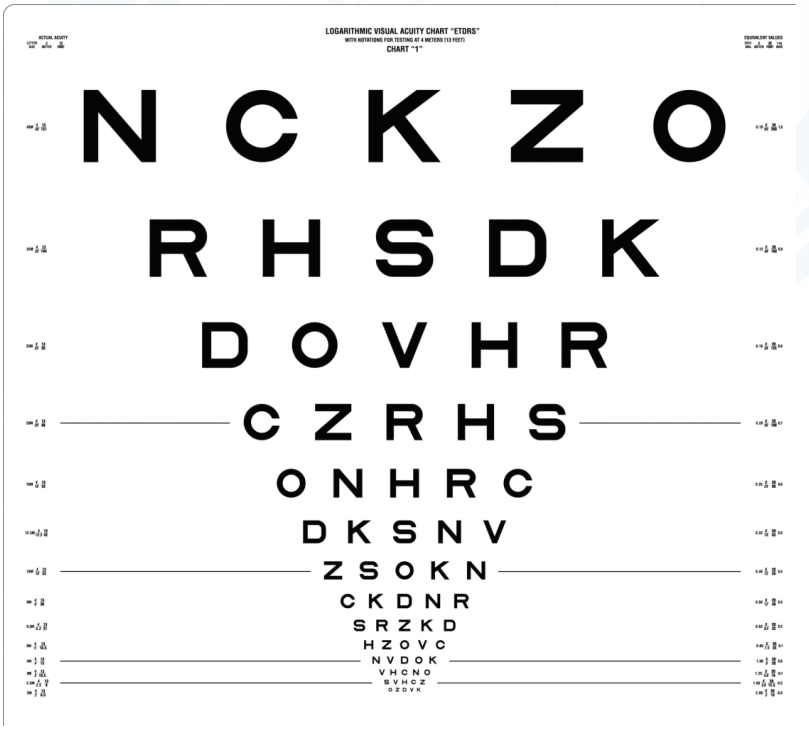
Active space

LOCAL SNELLEN'S CHART





MOST ACCURATE -ETDRS/ LOG MAR CHART



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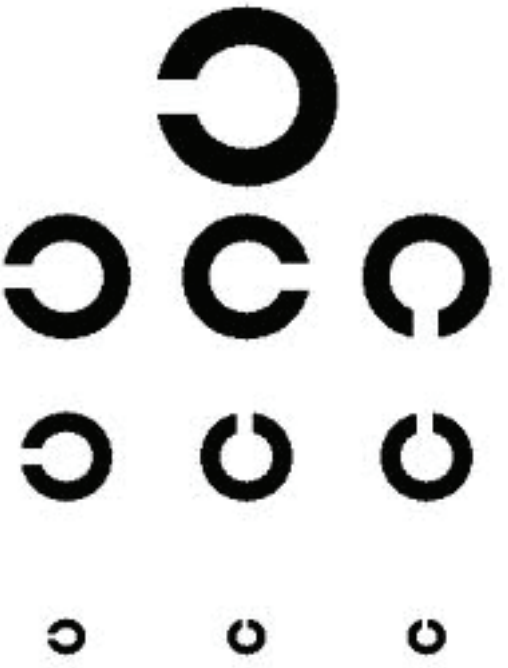
LANDOLT'S RING- ILLITERATES



PRESCHOOL CHARTS

- Allen charts
- Lea
- Animal
- Colourful charts

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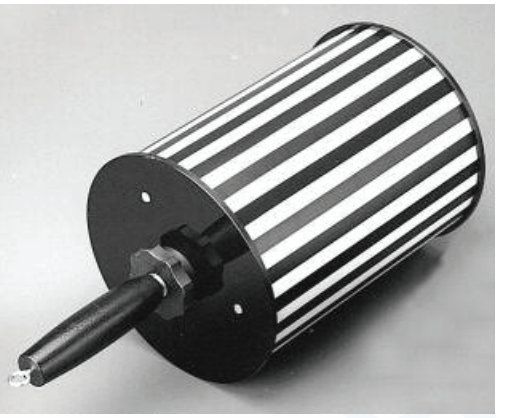
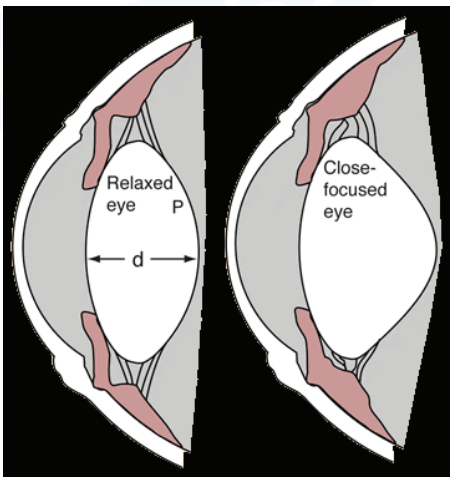
PREVERBAL/ UNCOOPERATIVE

- Optokinetic nystagmus



NEAR VISION

- Near rays are divergent, become parallel after travelling 6 metres
- Need 3 reflexes for seeing near vision
- Accommodation- change in shape of lens to increase power of lens to look at near objects
- Near visual impairment (NVI) defined as inability to read N6 at 40 cms
- NVI more damaging than distance vision



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TRIPLE RESPONSE/NEAR REFLEX/ ACCOMMODATION REFLEX

Accommodation –change in power lens by changing shape of lens to see objects at variable distances

- Blurred vision stimulates ciliary muscles contraction – zonules relax – lens becomes spherical –increases power
- Convergence – two eyes come closer –MR muscles
- Miosis – sphincter pupillae

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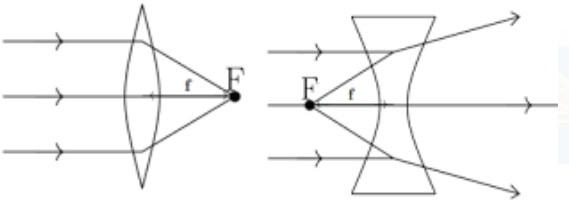
ACCOMMODATION DEFINITIONS

- Far point – farthest object that can be seen clearly without accommodation
- Far point of emmetrope – infinity
- Near point – nearest point that can be seen clearly with maximum accommodation
- Near point of emmetrope – 25 cms

SPHERICAL LENSES

- Convex lenses – causes convergence of rays + lenses
- Causes magnification – Each 1 D magnifies 2%
- Concave lenses – causes divergence of rays
- - lenses
- Causes minification – each 1 D minifies 2% size



Converging lens Diverging lens

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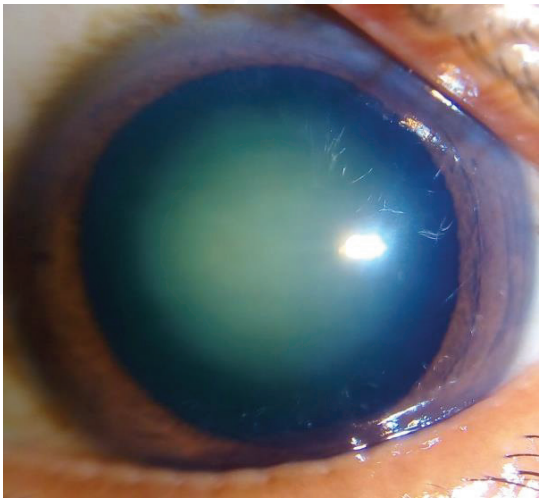
BINOCULAR SINGLE VISION

- Fusion of two images occurs in occipital cortex
- Fusion requires
- Bifoveal images
- Both images similar
- No fusion leads to -

CATARACT

- Opacification of lens or capsule
- Most common cause of blindness



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CAUSES OF CATARACT:

- Age-related (>55-60 years)
- Congenital/ Developmental
- Traumatic
- Complicated
- Metabolic
- Heat
- Radiation
- Drug-induced

AGE RELATED /SENILE CATARACT

- Exposure to UV light
- Smoking
- Exposure to heat , dust ,humidity and pollution
- Vitamin A , C, E protect against cataracts
- Women have more cataracts

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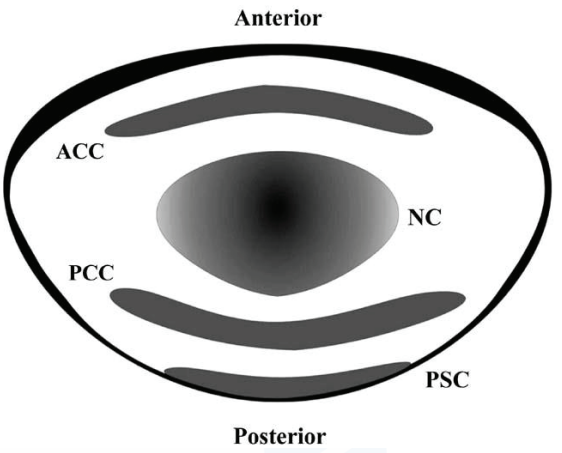
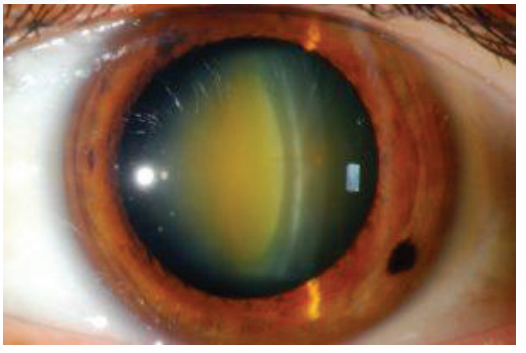
TYPES OF CATARACT:

- Nuclear
- Cortical
- Post. subcapsular



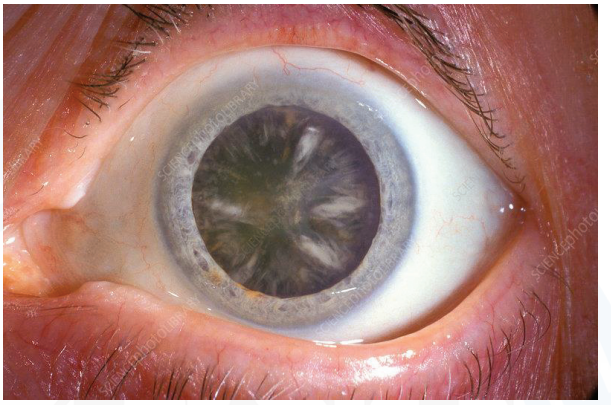
NUCLEAR CATARACT

- Decreased distant vision due to index myopia
- Hemeralopia – -blindness



CORTICAL CATARACT

- Cataract
- Nyctalopia



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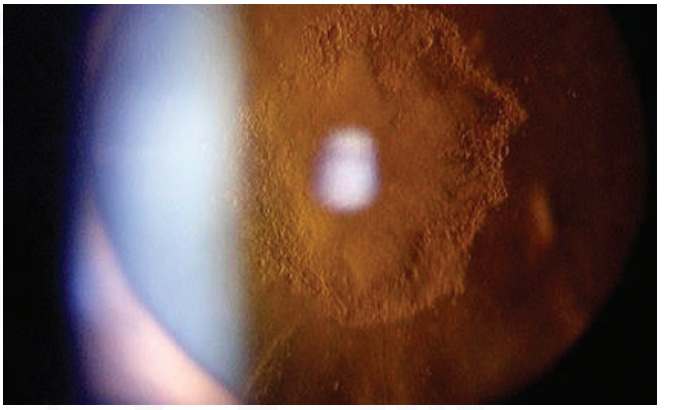
POSTERIOR SUBCAPSULAR

- Max Visual handicap
- Max Glare – closest to - Point
- More posterior the cataract – more drop in vision
- - vision impaired



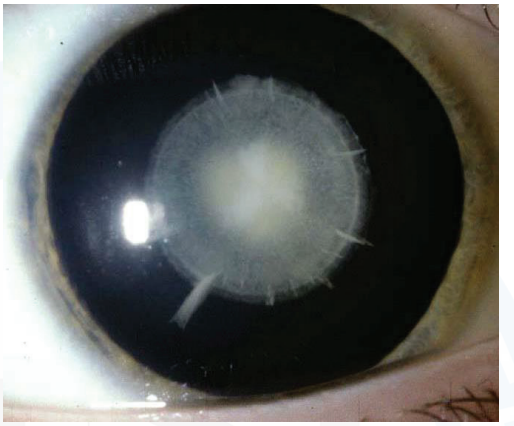
CONGENITAL/DEVELOPMENTAL CATARACT

- TORCH Infections
- Critical period for development of fixation: 2-4 months

Active space

CONGENITAL CATARACT

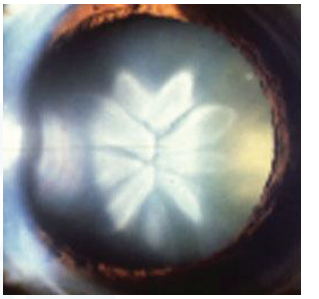
- M/C causing visual loss → Zonular/Lamellar/ - Cataract



Active space

TRAUMATIC CATARACT

- Blunt / penetrating Trauma
- Rosette Cataract- petals of a rose
- - ring



COMPLICATED CATARACT

- Secondary to intraocular disease
- Uveitis
- Glaucoma
- Retinitis Pigmentosa
- -

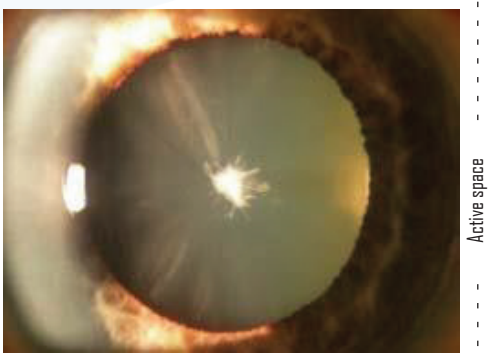
Active space

COMPLICATED CATARACT

- Post sub-capsular
- Bread-crumb appearance
- -

METABOLIC CATARACT

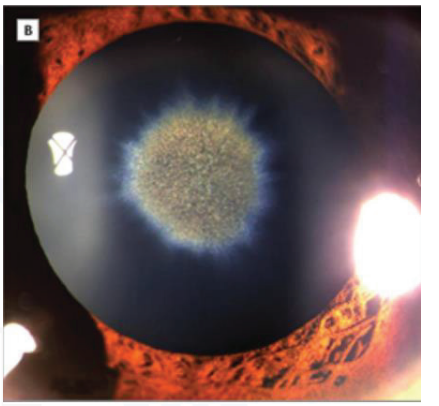
• Diabetes	
• Early senile	True diabetic/Snowflake
• Early to mid 40's	Younger
• Type 2 DM	Type 1 DM
• Gradual loss of vision	Acute onset
• PSC	- Shaped



Active space

METABOLIC CATARACT

- Wilson's - Sunflower
- Galactosemia – Oil drop
- Myotonic Dystrophy – Christmas tree



HEAT CATARACT

- True exfoliation cataract
- Glass-blower's cataract
- Iron-worker's cataract

Active space

RADIATION CATARACT

- Ionizing radiation – X rays most cataractogenic
- Posterior subcapsular cataract
- Lens – Most sensitive to radiation

DRUG-INDUCED CATARACT

- Systemic steroids – PSC cataract
- Amiodarone
- Busulphan
- Chlorpromazine/Chloroquine
- Dexamethasone

Active space

STAGES OF CATARACT

- 1. Incipient
- 2. Immature
- 3. Mature
- 4. Hypermature

INCIPIENT

- Earliest stage
- Mild blurring of vision
- Glare
- Loss of contrast sensitivity

Active space

IMMATURE

- Partially opaque
- Moderate blurring of vision
- Yellowish grey
- Iris shadow

MATURE

- Completely opaque
- Severe loss of vision
- White
- Intumescent – highest chance of glaucoma

Active space

HYPERMATURE

- Loss of volume – lens shrinks
- Milky white cataract
- Loss of zonular support → phacodonesis → Subluxation / dislocation

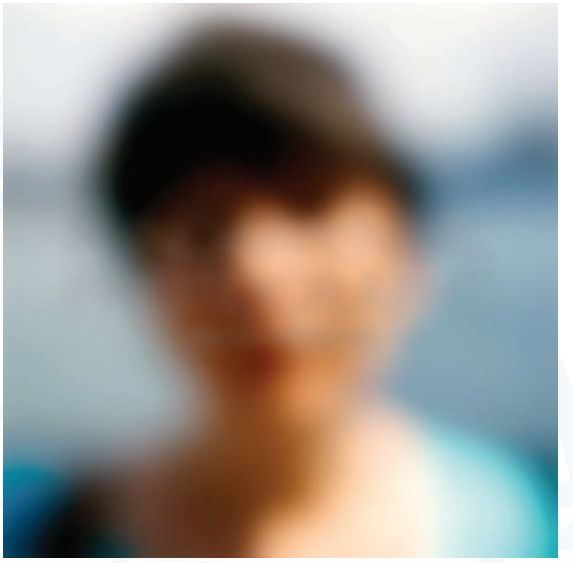
HYPERMATURE CATARACT

- **Morgagnian**
  - Cortex liquefies
  - Nucleus falls into it
- **Sclerotic**
  - Cataracta Brunescent- brown colour
  - Cataracta Nigra- hardest/black in colour

Active space

SYMPTOMS OF CATARACT

- Blurring of vision
- Diplopia/ Polyopia
- Coloured halos



SLIT LAMP- MOST IMPORTANT INSTRUMENT

Active space

TREATMENT

- Glasses
- Surgery

INDICATIONS OF SURGERY

- Visual handicap
- Advanced cataracts – Mature / Hypermature
- Avoidable in younger patients- loss of accommodation

SURGERY

1. ICCE (Intracapsular cataract extraction)
2. ECCE with PCIOL (Extracapsular cataract extraction with PCIOL)
3. Phacoemulsification

Active space

ICCE

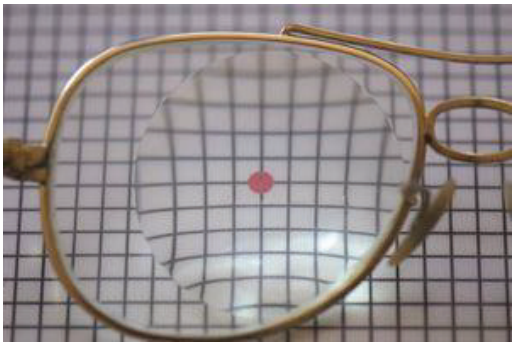
- Remove cataract + Capsule
- Aphakia
- Corrected by aphakic glasses
- Causes multiple problems

  Only indication ICCE → Subluxated/dislocated cataract

APHAKIA SIGNS /SYMPTOMS

- Diplopia
- Jack in the box scotoma / roving ring scotoma

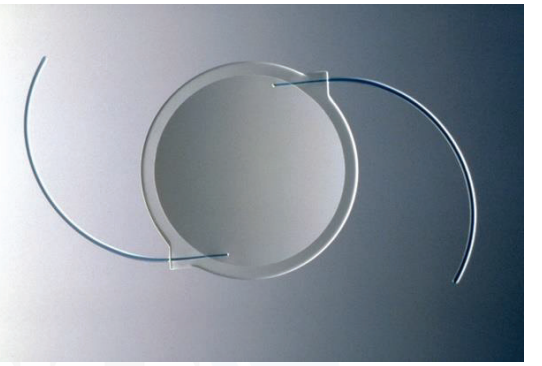
  ..... distortion
- Problems of alignment and orientation
- Deep AC



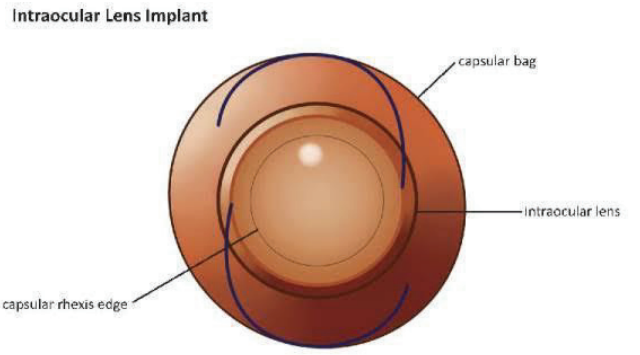
Active space

ECCE (WITH PCIOL)

- Cataract removed through capsulotomy , capsule left in situ
- Intraocular lens placed on posterior capsule
- Pseudophakia
- PMMA IOL



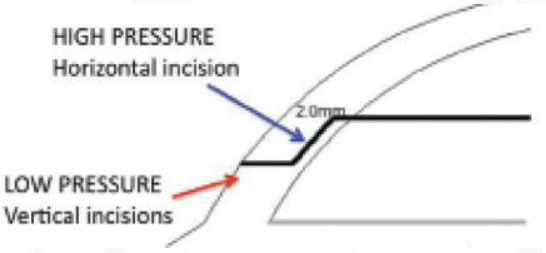
IOL IN THE CAPSULAR BAG



Active space

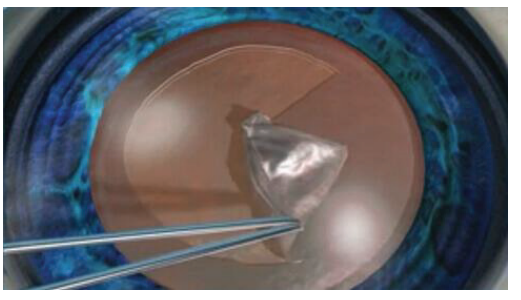
PHACOEMULSIFICATION

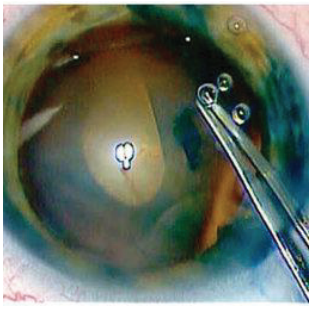
- Stitchless
- < 3mm incision
- Valvular/Multiplanar incision
- Self-sealing
- Cataract emulsified
- Foldable IOL
- Silicone/Acrylic IOL



ANTERIOR CAPSULOTOMY –CAPSULORHEXIS

- Central 5-6 mm capsulotomy –remove cataract
- Trypan blue dye injected in mature/Hypermature cataract to visualize capsule

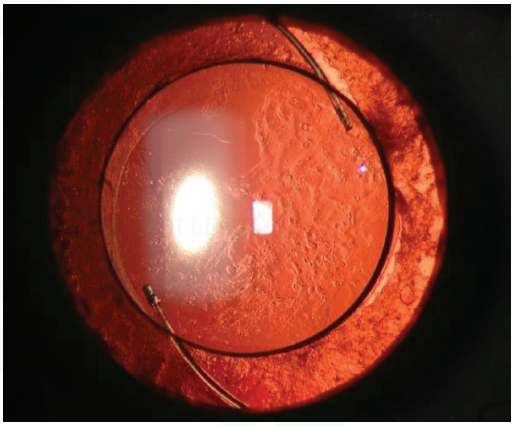




Active space

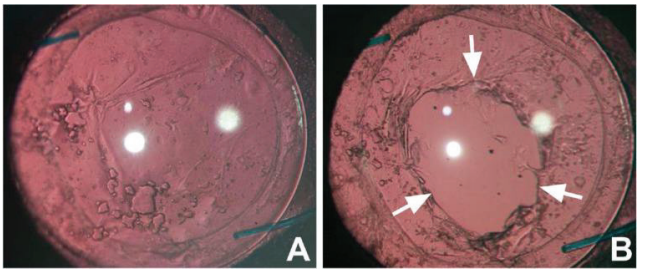
POST-OP COMPLICATIONS

- M/C -Secondary Cataract/After cataract/PCO
- due to posterior migration of LECs
- Two types
- Elschnig's pearls
- Soemmering's rings



MANAGEMENT -LASER CAPSULOTOMY

- Nd -YAG laser cuts posterior capsule
- Neodymium
- Yttrium
- Aluminium
- Garnet
- wavelength -1064 nm



Active space

POST-OP COMPLICATIONS

- Cystoid Macular Oedema(CME)
  - Painless blurring of vision
  - Metamorphopsia
  - Occurs in first two months
  - Irvine Gass Syndrome=Post cataract CME
  - Topical steroids / NSAIDS

ENDOPHTHALMITIS

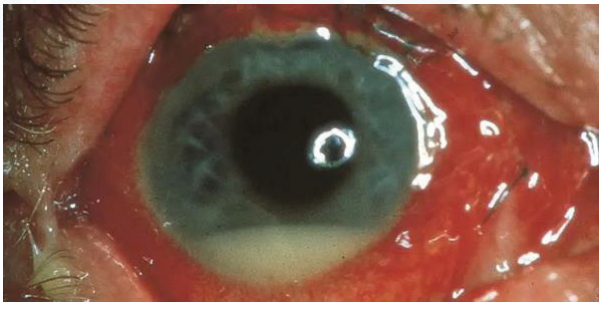
Purulent inflammation of the vitreous cavity

Early Onset (< 6weeks)

- Commonest: Staph epidermidis

Late Onset(>6 weeks)

- Commonest: Propionibacterium acnes



Active space

- Pain, redness, dramatic loss of vision
- DOC: Intravitreal Vancomycin, Ceftazidime
- -
- Most important precaution: Eyelid and eyelash cleansing with Povidone iodine

AMBLYOPIA

- Loss of vision in a child < 8 years age which has no obvious cause and cannot be corrected by optical or surgical means

Active space

CAUSES OF AMBLYOPIA

- Strabismus (40%) -m/c
- Refractive – uncorrected hypermetropia
- Form deprivation – least common, most severe
- visual axis obstructed – cataracts, ptosis, corneal opacities

CROWDING PHENOMENA

- In amblyopia stimuli presented in a crowd cannot be seen , but is visible as a single stimulus

Active space

TREATMENT OF AMBLYOPIA

- Occlusion therapy – 6-8 hours a day
- Penalization – penalise good eye with atropine

UVEITIS

Active space

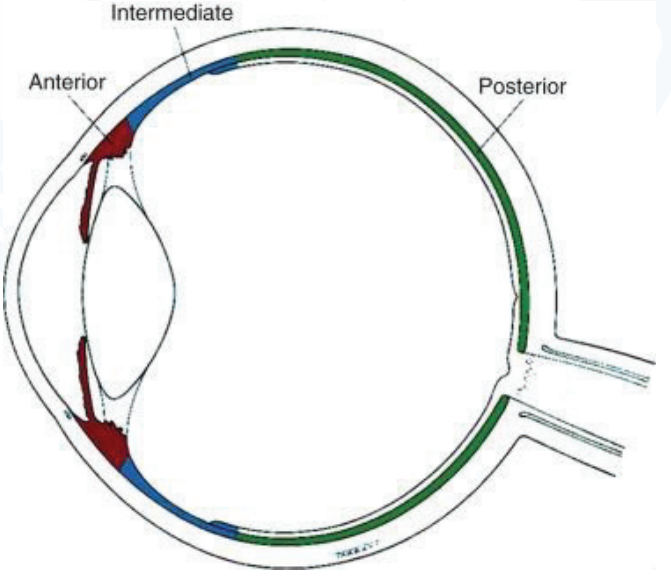
CLASSIFICATION

1. Anterior Uveitis
   - Iritis
   - Iridocyclitis (only pars plicata)

2. Intermediate uveitis
   - – Pars Planitis
   - – Vitritis

3. Posterior uveitis
   - – Choroiditis

4. Panuveitis
   - – Sympathetic Ophthalmitis



Active space

CLASSIFICATION OF UVEITIS

- Acute – uveitis lasting < 3 months
- Chronic - uveitis lasting > 3 months
- Granulomatous uveitis – caused by diseases forming granulomas – Tuberculosis , syphilis , leprosy , sarcoidosis , Lyme disease
- M/c /c granulomatous uveitis – sarcoidosis
- Non granulomatous uveitis – no granulomas

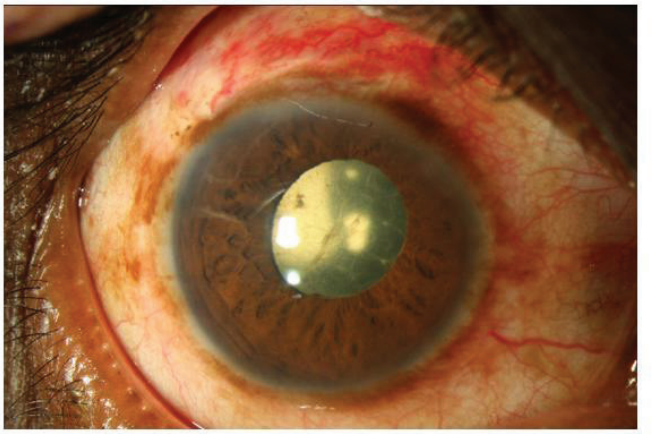
ANTERIOR UVEITIS – CAUSES

- Idiopathic
- HLA – B 27 Spondyloarthropathies
- Ankylosing spondylitis
- Inflammatory Bowel Disease- Crohn's/Ulcerative C
- Psoriatic arthritis
- Reactive Arthritis
- Conjunctivitis , Urethritis , aRthritis
- JRA/ Juvenile Idiopathic Arthritis
- Pauciarticular , ANA +, Rheumatoid Factor -

Active space

JRA /JIA

- M/c /c anterior uveitis in children
- IOL contraindicated in JRA



ANTERIOR UVEITIS

- Symptoms
  - Redness, pain, blurring of vision

Active space

SIGNS -CIRCUMCILIARY / CIRCUMCORNEAL CONGESTION

- Around ciliary body
- Reddish -violet colour

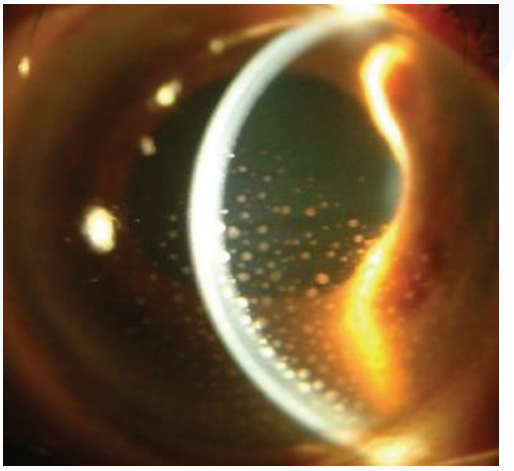
ANTERIOR UVEITIS

- Cells -WBC in AC
- Sign of activity

Active space

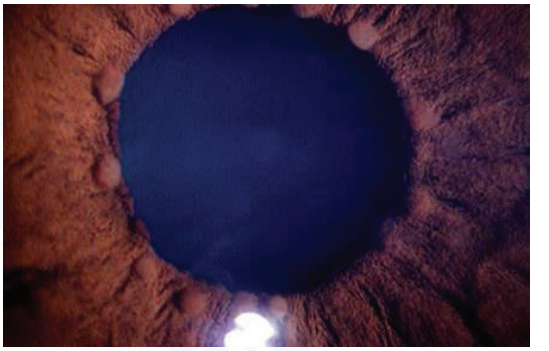
- Flare
  - Protein deposition in aqueous humor
  - Earliest sign of uveitis

- Keratic precipitates
  - Neutrophils & lymphocytes on corneal endothelium
  - Arlt's triangle – inferior
  - Mutton-fat KPs – granulomatous uveitis



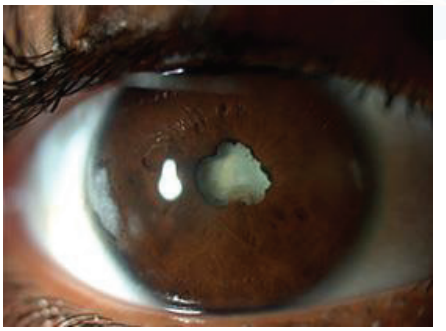
Active space

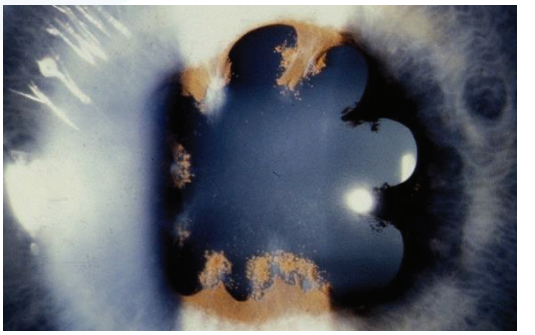
- Iris Nodules
  - Koeppe's- pupillary margin
  - Busaca's- surface of iris



- Synechiae
  - Anterior-Glaucoma
  - Posterior-Cataract
  - Festooned pupil –irregular

- Miosis
- Low IOP

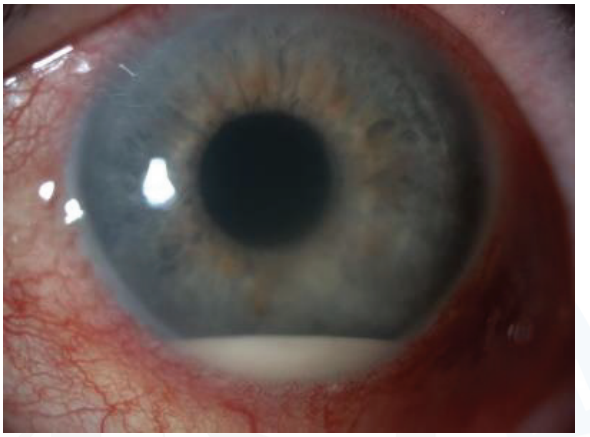




Active space

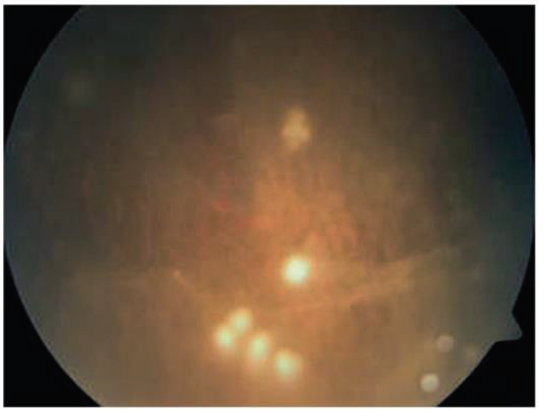
HYPOPYON

- Collection of WBC 's in AC
- Sign of severe uveitis



INTERMEDIATE UVEITIS

- Pars planitis with vitritis
- Idiopathic
- Sarcoidosis
- M/c - Floaters /Muscae
- CME –m/c/c loss of vision
- Snowballs and snowbanks



Active space

POSTERIOR UVEITIS

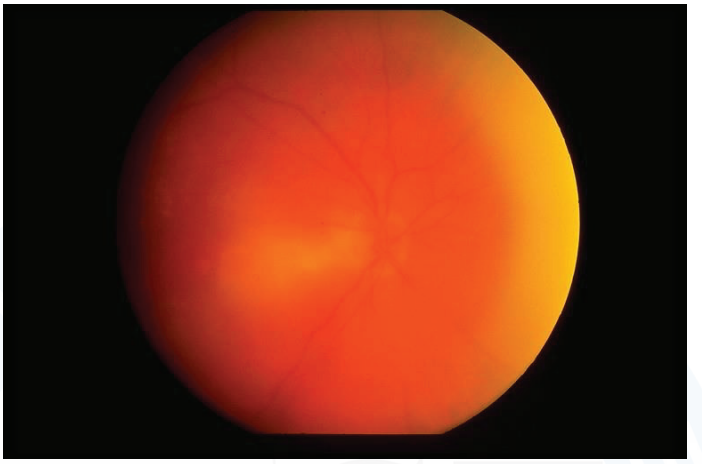
Infections	Non infectious
- Toxoplasmosis	- Sarcoidosis
- Tuberculosis	
- Toxocariasis	

POSTERIOR UVEITIS

- Painless blurring of vision
- Choroiditis
- Chorioretinitis
- Vitritis

Active space

HEAD LIGHT IN FOG



TREATMENT

- Anterior Uveitis
- Topical steroids DOC
- M/c side effect — Glaucoma
- Systemic steroids — Cataract

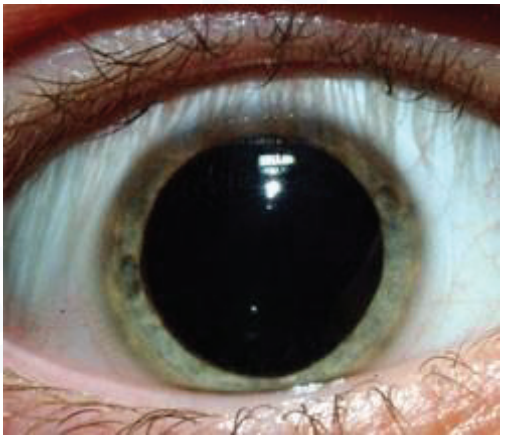
Active space

Cycloplegics

- Atropine 14 days
- Homatropine 3 days
- Cyclopentolate 1 day
- Tropicamide 6 hours

CYCLOPLEGICS

- Relax ciliary spasm
- Dilate pupil , break posterior synechiae



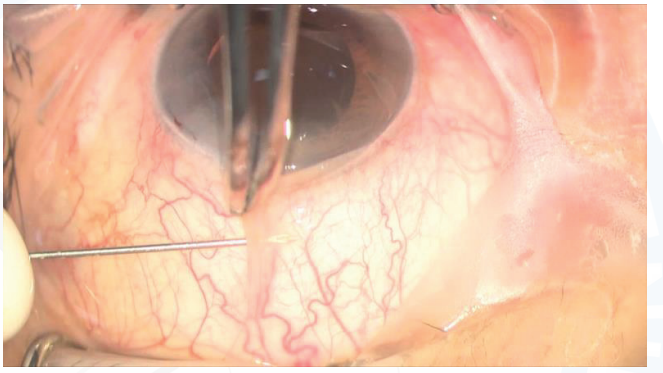
Active space

- In children – Atropine cycloplegic of choice
- Atropine drops NOT used , only ointments
- Systemic absorption through nasolacrimal duct -80%

TREATMENT INTERMEDIATE UVEITIS

- Steroid Injections -Triamcinolone
  - Subconjunctival
  - Subtenon's

- Cycloplegic not required



- Posterior uveitis
- Antimicrobials for infectious disease
- DOC of Toxoplasmosis during pregnancy -
- Systemic steroids for non infectious causes

Active space

SUMMARY

Anterior uveitis	Intermediate uveitis	Posterior uveitis
Iridocyclitis	Pars planitis & vitritis	Chorioretinitis
Idiopathic	Idiopathic	Toxoplasmosis
Pain ✓ Redness ✓ Loss of vision ✓	Pain ✖ Redness ✖ Loss of vision ✓ Muscae/floaters	Pain ✖ Redness ✖ Loss of vision ✓
Cells	Snowballs & snowbanks	Toxoplasmosis – Headlight in fog Vitritis, vasculitis, chorioretinitis
Topicals steroids Cycloplegics	Inj Triamcinolone	Systemic steroids Antimicrobials

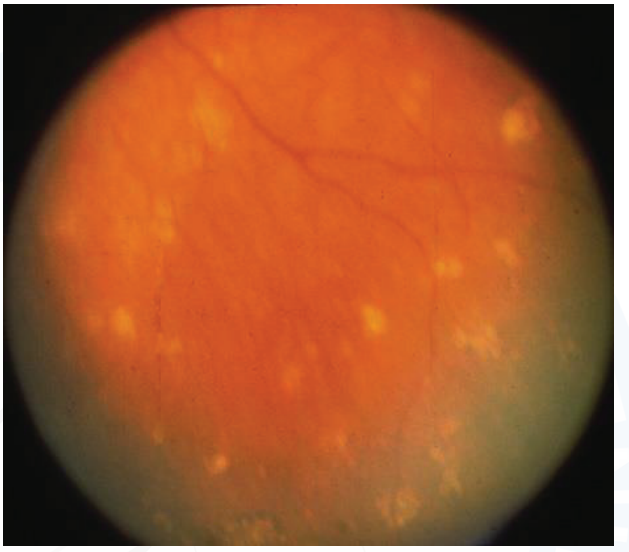
PAN UVEITIS -SYMPATHETIC OPHTHALMITIS

- B/L granulomatous panuveitis following trauma
- Penetrating/perforating injury
- Injury to ciliary body highest chance of SO
- Surgical repair > 48 hours after injury
- 2 weeks – 3 months – commonest presentation
- Blurred vision , photophobia ,pain
- AC reaction- cells , flare , KP's , posterior synechiae
- Vitritis , papillitis , retinal detachment

Active space

SYMPATHETIC OPHTHALMITIS

- Dalen Fuchs' Nodules
- M/C seen in children



Earliest Symptom

Photophobia

Accommodation failure

Active space

Earliest Sign

Retrolental Flare

Mutton fat KPs

- Treatment
  - ➢ Enucleation
  - ➢ Repair the injury
  - ➢ Systemic steroids

OCULAR HIV

- Commonest manifestation: Cotton wool spots/ soft exudates (retinal microangiopathy)

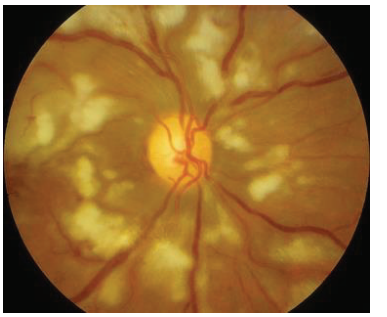
  - M/c infection:
  - M/c /c loss of vision

- Pizza pie /scrambled eggs and ketchup

- DOC -

  M/c tumour Kaposi Sarcoma

Active space



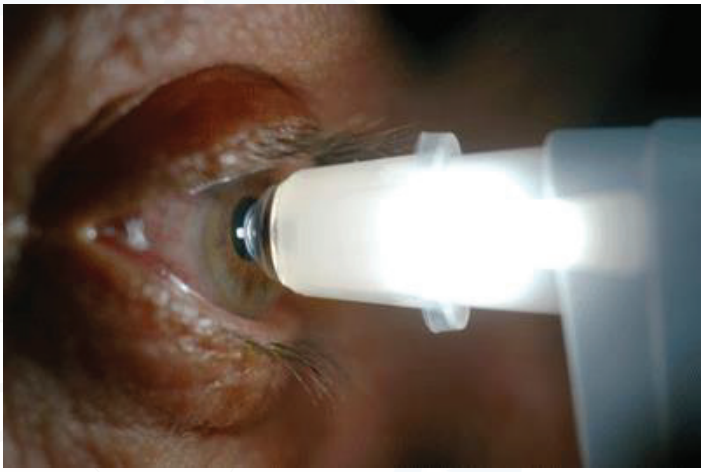
INTRAOCULAR PRESSURE IOP/IOT

- Pressure of aqueous / vitreous on cornea/sclera
- IOP varies from 10 -21 mm of Hg
- Mean IOP =16 mm, > 21 mm of Hg glaucoma
- Measured by tonometer - gold standard is Goldmann Applanation Tonometry
- Depends on Pachymetry

Active space

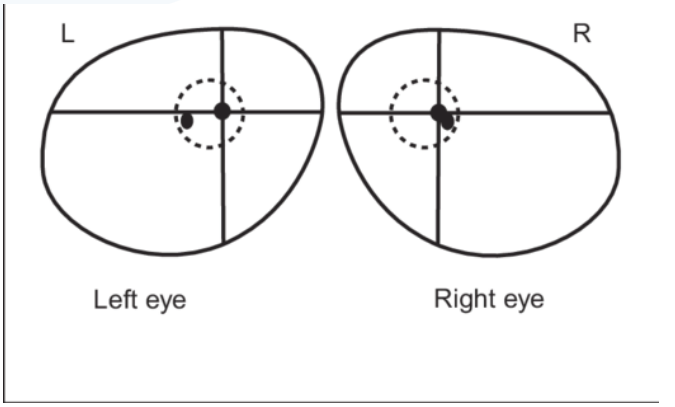
CENTRAL CORNEAL THICKNESS

- Measured by pachymetry
- Average CCT – 540 microns
- Thicker corneas measure falsely high IOP
- Thinner corneas measure falsely low IOP



VISUAL FIELD

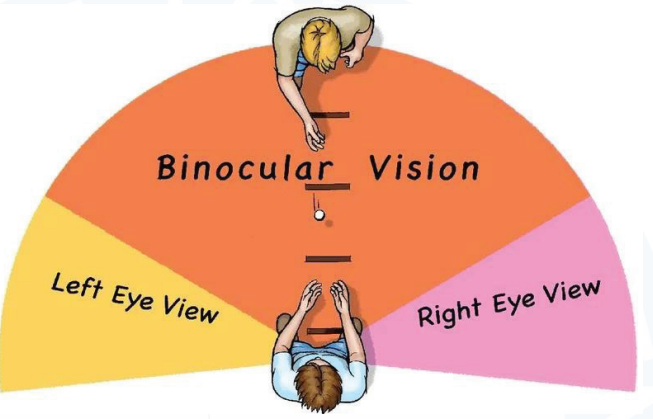
- An area of space which is visible without moving the gaze from a central target



Active space

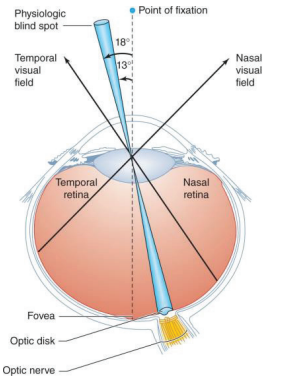
MONOCULAR FIELD

- Superior field 60 degrees
- Nasal field 60 degrees
- Inferior field 70 degrees
- Temporal field 100 degrees
- Binocular field covers 180 degrees



RELATIONSHIP OF VISUAL FIELD TO RETINA

- Inverse relationship – visual field - opposite retina
- Visual field measures -
- Visual field measured by -
- Humphrey's Visual field analyser most accepted

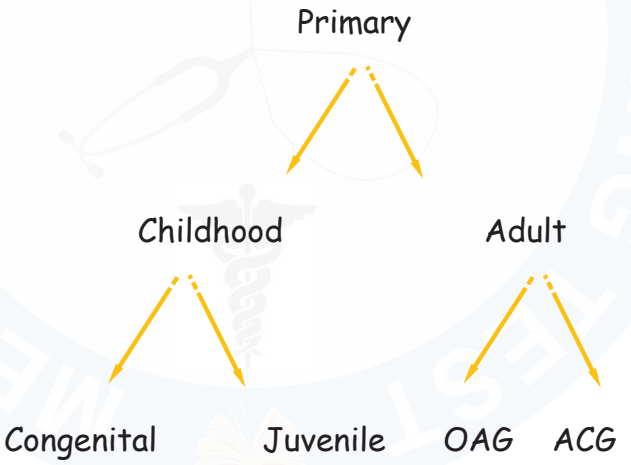


Active space

GLAUCOMA

1. Increased IOP
2. Optic Disc damage
3. Visual field defects

• Primary	• Secondary
• cause of IOP unknown	• Uveitis - Neovascular - Trauma - Steroid induced glaucoma



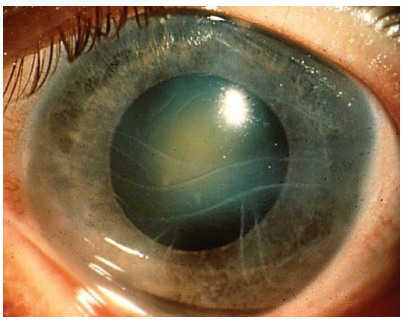
Active space

CONGENITAL GLAUCOMA

- Buphthalmos
- Congenital anomaly -Barkan's Memb blocks angle
- Autosomal recessive
- Consanguineous marriage

- Triad:
  1. Lacrimation
  2. Photophobia
  3. Blepharospasm

- Hazy Cornea- corneal edema
- Enlarged eyeball
- Axial myopia
- Haab 's striae

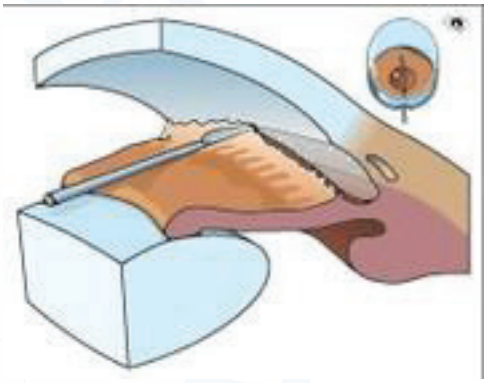


Active space

TREATMENT

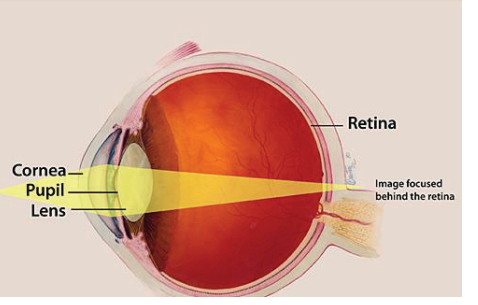
1. Goniotomy
   - Safer but rare
   - Hazy cornea-Gonioscopy difficult

2. Trabeculotomy- possible in hazy corneas



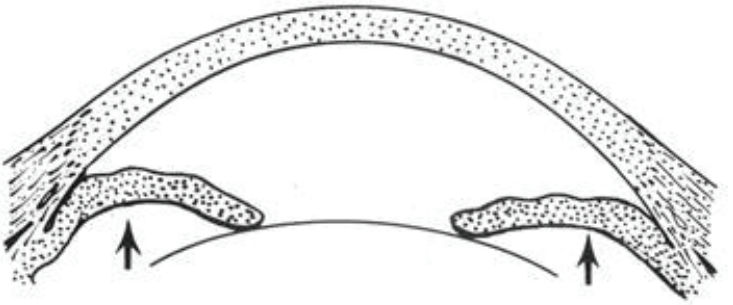
PREDISPOSING CAUSES -ANGLE CLOSURE GLAUCOMA

- Seen in
  - Middle aged women
  - Hypermetropia
  - -



Active space

MID – DILATED PUPIL □PUPILLARY BLOCK



ACUTE ANGLE CLOSURE GLAUCOMA

- Late night attack
- Mid-dilated pupil
- Pupillary block
- Angle closure
- Sudden rise in IOP

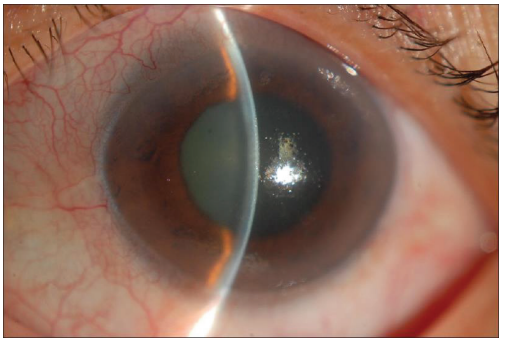
Active space

- Symptoms:

  Severe pain Vomiting

  - Redness
  - -
  - Sudden drop in vision

- Signs:
  - Stony hard eye
  - Shallow AC
  - -
  - Vertically oval, mid-dilated, non reacting to light



Active space

Fincham's Test

Colored Haloes

Cataract

Split up & unite again

ACG

Do not split(Stenopic slit)

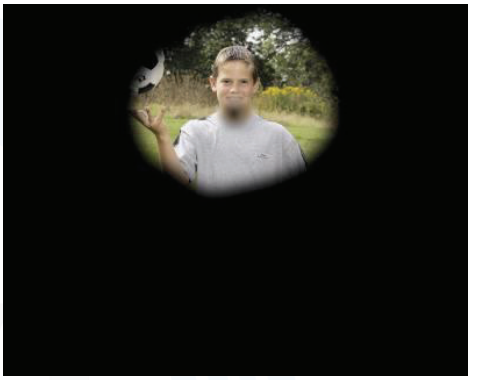
OPEN ANGLE GLAUCOMA

- Colored races
- Myopic
- Middle aged
- -
- Pathology – Trabecular meshwork fibrosis

Active space

OAG – SIGNS / SYMPTOMS

- Slow, progressive rise of IOP over many years No pain / no corneal oedema/ no redness / no coloured halos /no blurring of vision Finally develops tubular/ tunnel vision



ONLY SYMPTOM OF OAG

- Frequent change of presbyopic glasses

Active space

D/D OF FREQUENT CHANGE OF GLASSES

- Frequent change of glasses in old patients-cataract
- Frequent change of glasses in young patient-Keratoconus

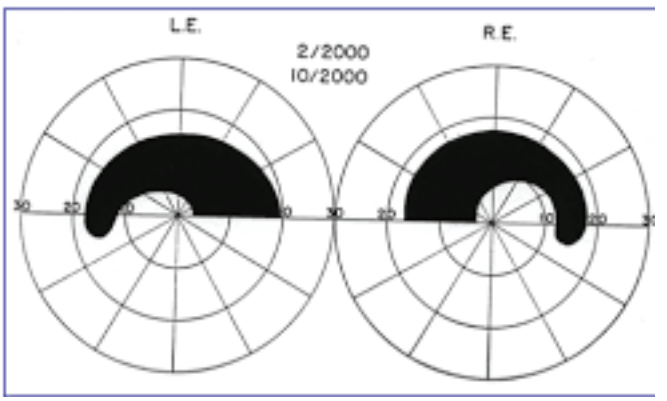
ACG	OAG
• Female • Middle aged • Hypermetropic • Colored haloes, ↓ VA • Pupillary Block	• No gender predisp • Middle aged • Myopic • No symptoms • Trabecular Fibrosis

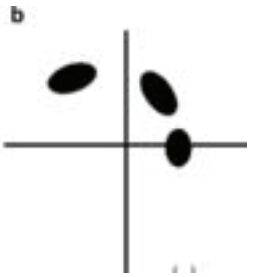
Active space

PERIMETRY

VISUAL FIELD DEFECTS OF GLAUCOMA

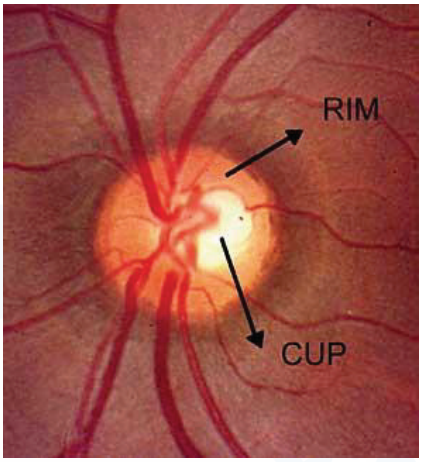
- Paracentral scotoma
- Bjerrum's scotoma
- Nasal step
- Arcuate scotoma





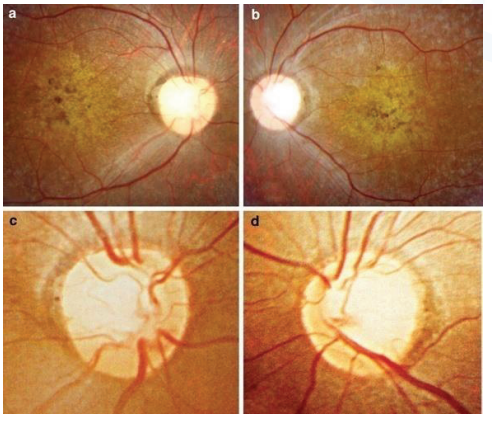
Active space

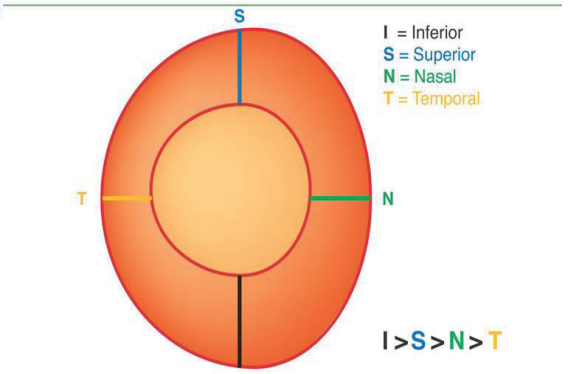
NORMAL OPTIC DISC



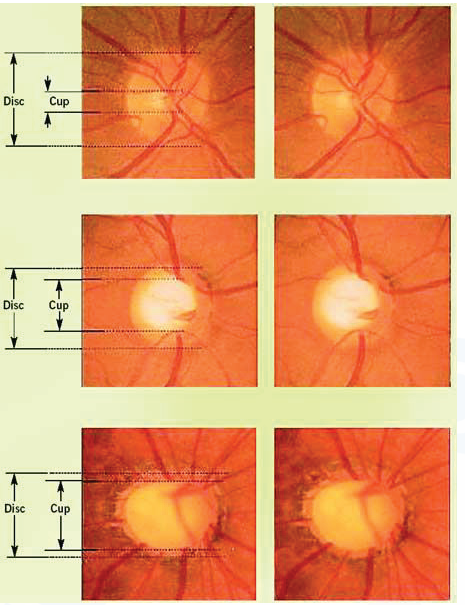
OPTIC DISC CHANGES IN GLAUCOMA

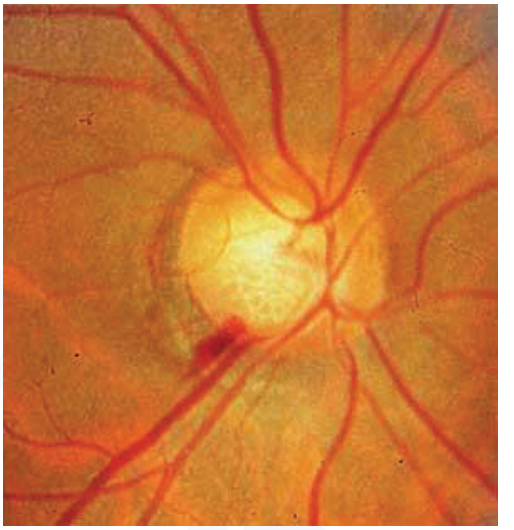
- Increased CDR
- Disc Pallor
- Splinter hemorrhage
- Sign of nasalization
- ISNT rule broken





Active space





GLAUCOMATOUS OPTIC ATROPHY

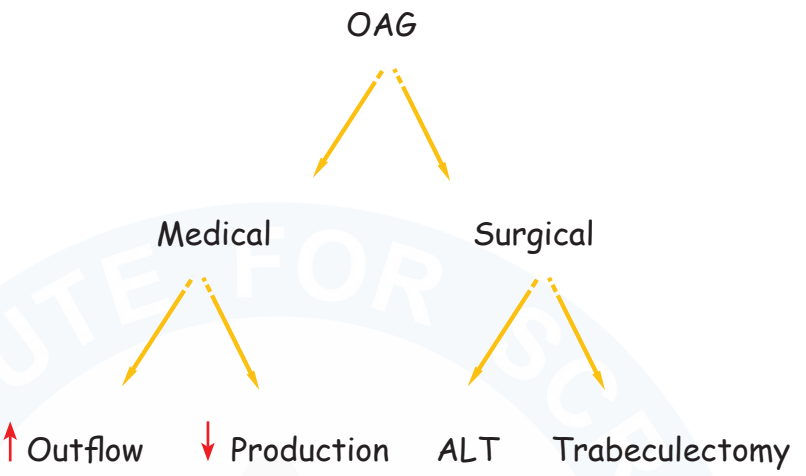
Active space

DIRECT OPHTHALMOSCOPE

- Examination central retina –disc , fovea , macula
- Virtual , erect , magnified image
- Magnification X 15
- Close to patient

Active space

TREATMENT



OAG TREATMENT

1. Medical
2. Argon Laser Trabeculoplasty
3. Trabeculectomy

Active space

Cholinergic Agonists(Miotics)

- Pilocarpine/Carbachol
  - Increases outflow

MIOTICS

- Side Effects:
  - Uveitis
  - Ciliary spasm
  - Myopia
  - Retinal Detachment

Active space

ß – BLOCKERS

- Decrease production of aqueous
- Non-selective Timolol, Levobunolol
- Selective-Betaxolol

Side Effects:

- Contraindicated in bronchial asthma/Arrythmias/COPD

CARBONIC ANHYDRASE INHIBITORS

- Decrease production
- Systemic: Acetazolamide/ Methazolamide
- Topical: Dorzolamide, Brinzolamide

Active space

ACETAZOLAMIDE/METHAZOLAMIDE

SIDE EFFECTS:

- C/I Sulfa-Allergy
- Hypokalemia
- Acidosis
- C/I Kidney stones
- C/I CRF
- C/I Hepatic failure

DORZOLAMIDE/BRINZOLAMIDE

- Topical CAI drugs
- Safest anti glaucoma drugs
- DOC in children
- S/E: corneal decompensation

Active space

α2–AGONIST

Reduces aqueous secretion / ↑ Uveo scleral outflow

- Brimonidine
- C/I in children
- S/E: drowsiness, depression
- Neuroprotection

PG ANALOGUE

Latanoprost/Bimatoprost/Tafluprost

- Increases outflow – uveoscleral pathway
- DOC for OAG and Normal tension glaucoma
- Most potent anti-glaucoma drug

PGA

Side Effects:

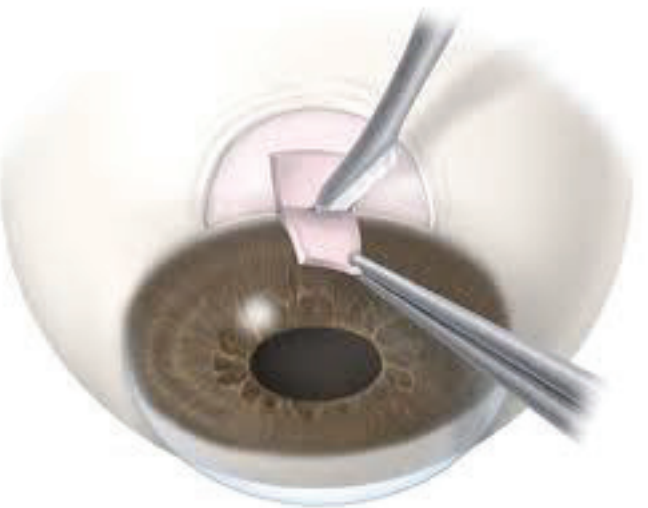
- Uveitis
- Cystoid macular oedema
- Iris hyperchromia
- Blephroconjunctivitis
- Trichomegaly

Active space

HYPEROSMOTICS

- Mannitol- intravenous , short – acting
- DOC: Acute ACG
- S/E: Decompensation in case of CHF
- Glycerol- oral sweet syrup
- C/I: DM

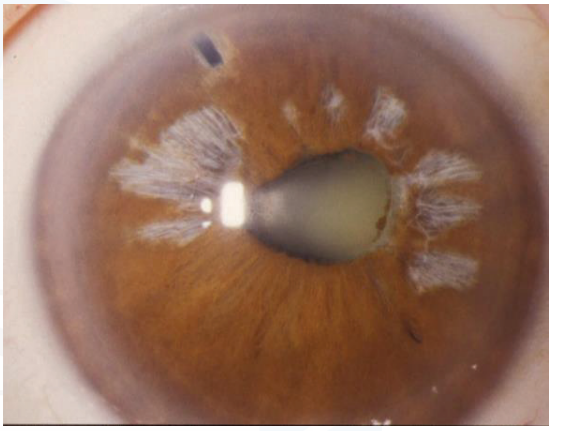
TRABECULECTOMY



Active space

ACG MANAGEMENT

- Immediate -Injection Mannitol /acetazolamide Final -Peripheral Iridectomy/ Laser Iridotomy
- Prophylactic Nd YAG Laser Iridotomy in fellow eye
- If LI fails , then trabeculectomy



NEOVASCULAR GLAUCOMA

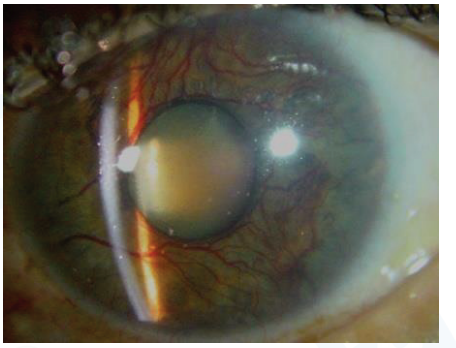
Causes:

- Diabetes
- CRVO
- Ocular Ischemic Disease

Active space

PATHOLOGY OF NVG

- Retina - hypoxia - VEGF - Neovascularization - rubeosis iridis (NVI)- zipper like angle closure -NVG
- Secondary angle closure glaucoma



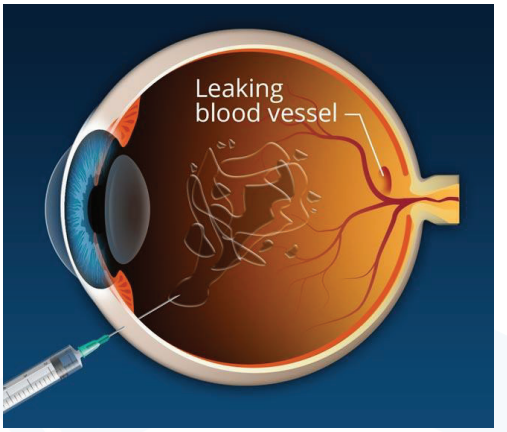
NEOVASCULAR GLAUCOMA

Early

- Stop neovascularization
  - Anti VEGF drugs
  - Bevacizumab/ Ranibizumab
  - Treat retinal hypoxia - Pan Retinal Photocoagulation

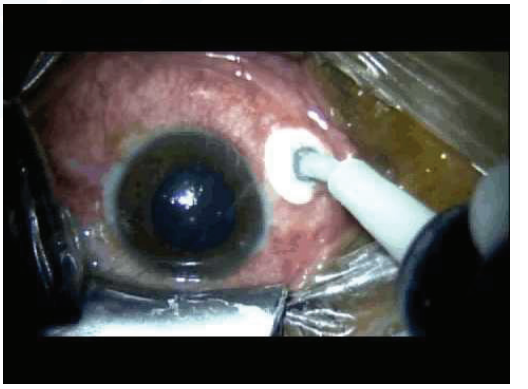
- Reduce IOP
  - Drugs- Latanoprost and Pilocarpine C/I
  - Trabeculectomy

Active space



LATE GLAUCOMA

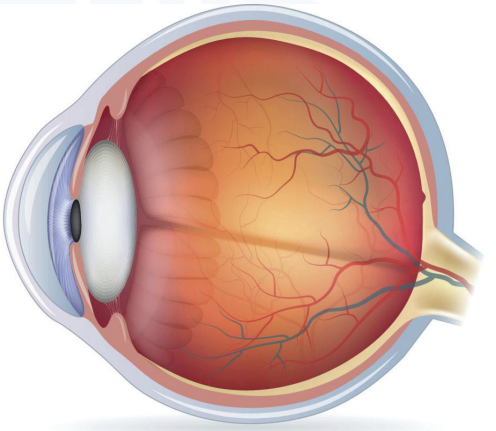
- Absolute glaucoma
- Cyclodestruction
- Diode laser cyclophotocoagulation (DLCP)
- Cyclocryopexy



Active space

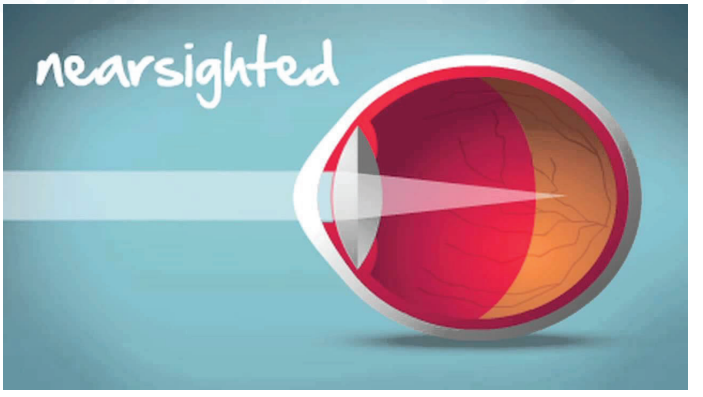
REFRACTION AND OPTICS

- Depends upon
- Axial length
- Corneal curvature
- Lens thickness
- AC depth



MYOPIA

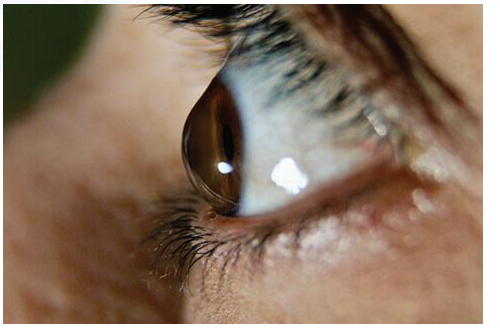
- Rays of light focus in front of retina
- Myopia term derived from half closed eyes



Active space

TYPES OF MYOPIA

- Axial myopia – longer eyeball , each 1 mm extra length equates to 3 Dioptres of myopia
- Curvature myopia – increased curvature of cornea – keratoconus



- Index myopia – increased refractive index – nuclear cataract

Active space

SIGNS/SYMPTOMS

- Blurring of distant vision
- Can see near objects
- Headaches

CORRECTION OF MYOPIA

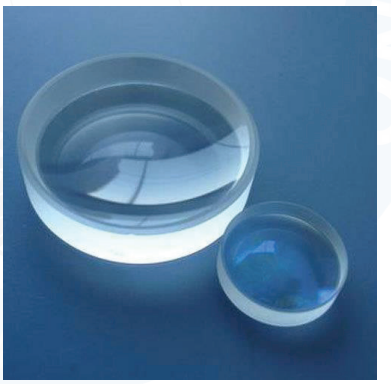
Concave lenses correct myopia

Wears glasses close to eyes

Concave lens causes minification

Myopes should be undercorrected

Duochrome test for under correction

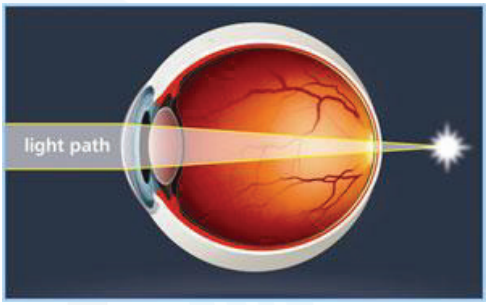


MIST

Active space

HYPERMETROPIA / HYPEROPIA

- Parallel rays from infinity focus behind retina
- Axial hyperopia –most common – each 1 mm of shortening causes 3 D hyperopia
- Newborns are hypermetropic
- At birth 2-3 Dioptres of hyperopia
- Children develop 6/6 by 6-8 years of age



SIGNS/SYMPTOMS

- Headaches / eye strain – asthenopia
- Asthenopia more for near than distance
- Uncorrected hypermetropia leads to
- Amblyopia
- Early presbyopia convergent squint

Corrected by convex lenses – glasses worn lower on the nose

Each 1 dioptre causes 2 % magnification

Uncorrected myopia my lead to divergent squint

Active space

CYCLOPLEGIC REFRACTION

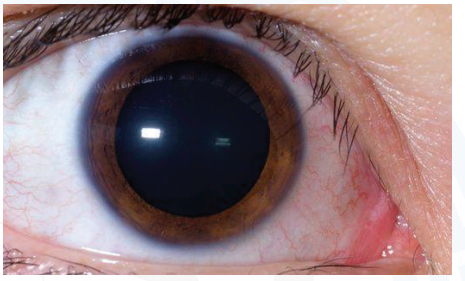
Relax ciliary muscles in abnormally strong accommodation

Hypermetropia

Children (Atropine - upto 7 years)

Homatropine - (7 to 10 years)

Convergent squint

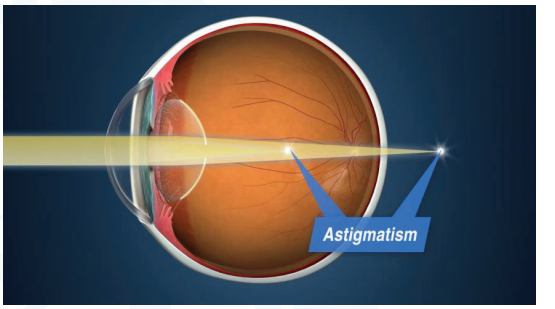


MYOPIA	HYPERMETROPIA
Rays focus in front of retina retina	Rays focus behind
Blurring of distance vision	Asthenopia
Concave lenses	Convex lenses
Worn close to eyes	Worn on tip of nose
Minification	Magnification
Poor accommodation	strong accommodation
Uncorrected –divergent squint	convergent squint

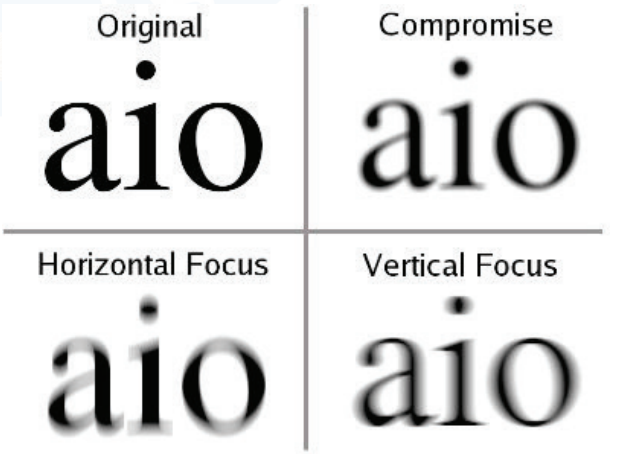
Active space

ASTIGMATISM

- Condition where there are two different foci
- Due to non – uniform curvature of cornea
- Most common refractive error
- Corrected by cylindrical/toric lenses
- Maximum asthenopia



VISUAL QUALITY IN ASTIGMATISM



Active space

READING VISION ASTIGMATISM

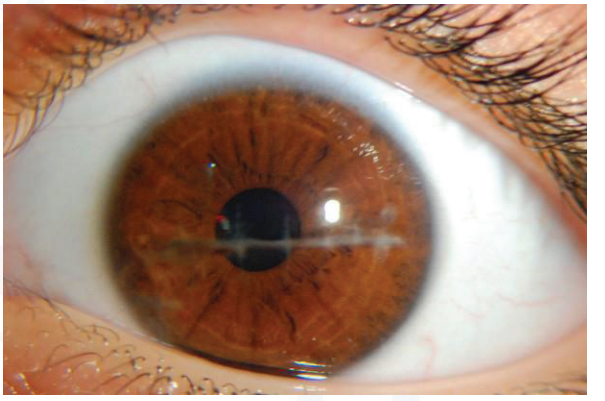
SYMPTOMS

- Asthenopia
- Monocular diplopia
- Distortion
- Headaches
- Ocular fatigue
- Halo around lights

Active space

REGULAR /IRREGULAR ASTIGMATISM

- Regular - corneal surface smooth -principal meridians(steepest/ flattest corneal curvatures) are perpendicular to each other
- Irregular astigmatism - corneal surface irregular - Pterygium , keratoconus , post - traumatic scarring
- Only contact lens can correct



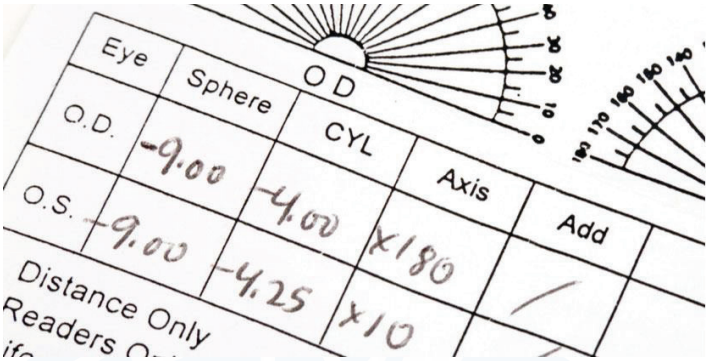
TYPES OF REGULAR ASTIGMATISM

- Simple myopic astigmatism
- Simple hypermetropic astigmatism
- Compound myopic astigmatism
- Compound hypermetropic astigmatism
- Mixed astigmatism

Active space

Date:

	SPHERE	CYLINDER	AXIS	PRISM	BASE
R	−0.50	+4.00	X 90		
L	−0.25	+4.25	X 75		



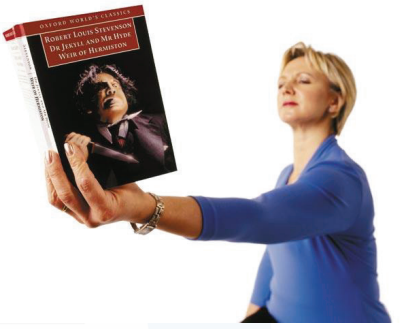
PRESBYOPIA

- Loss of accommodation with age
- Begins around 40 years of age
- Due to hardening of lens , fibrosis of capsule , weakening of ciliary muscles

Active space

SYMPTOMS OF PRESBYOPIA

- Near vision activities –reading , writing , knitting , cooking becomes blurred
- Holding reading materials at arm's length
- Fatigue after doing close work



PRESBYOPIC VISION

Active space

WITH CORRECTION

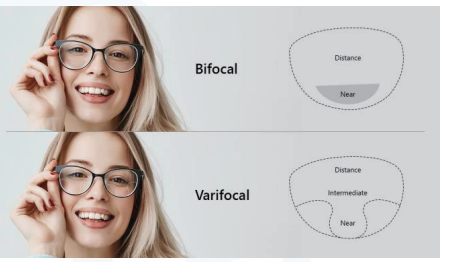
AGE RELATED PRESBYOPIC ADDITION

Age in years	Addition
40	1.00 D
45	1.50 D
50	2.00 D
55	2.50 D
60	3.00 D

Active space

CORRECTION OF PRESBYOPIA

- Convex lenses
- Bifocals for refractive error
- Progressives –a trifocal
- Top segment – distance
- Middle segment – intermediate
- Bottom segment – near



DARK ROOM PROCEDURES

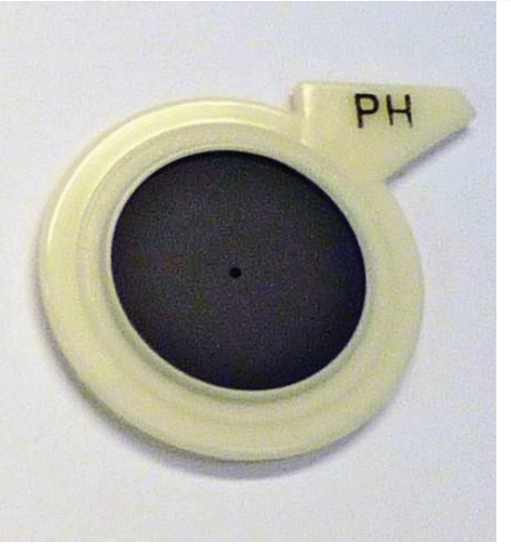
Active space

DARK ROOM PROCEDURES

- Retinoscopy
- Distant Direct Ophthalmoscopy
- Direct Ophthalmoscopy
- Indirect Ophthalmoscopy

PIN HOLE

- A plastic disc with a central hole, with varying diameters around 1 mm
- Vision improves with pin hole – refractive error
- Vision does not improve – organic error
- VA improves with pin hole , retinoscopy done to identify refractive error

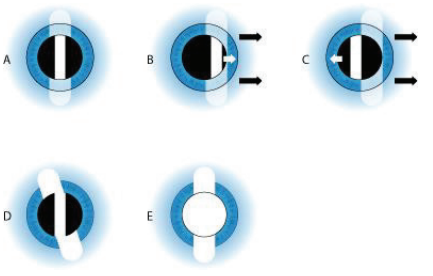


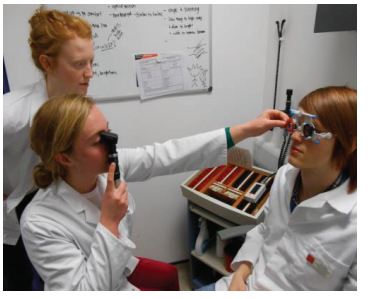
Active space

RETINOSCOPE

RETINOSCOPY

- Objective measurement of refractive error
- Estimates all refractive errors – myopia / hypermetropia / astigmatism
- Performed at 1 metre away from patient
- SPAM - same plus against minus
- Subtract for 1 metre distance

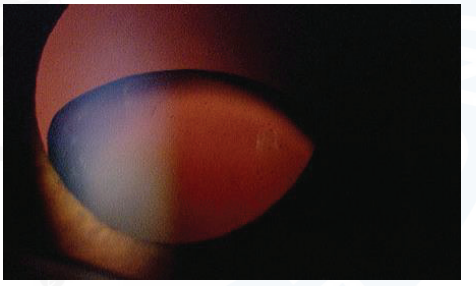




Active space

DISTANT DIRECT OPHTHALMOSCOPY

- Direct ophthalmoscope , at 25 cms
- Red glow – normal
- Grey glow – retinal detachment
- No glow – vitreous haemorrhage
- Dark crescent – subluxated lens



Active space

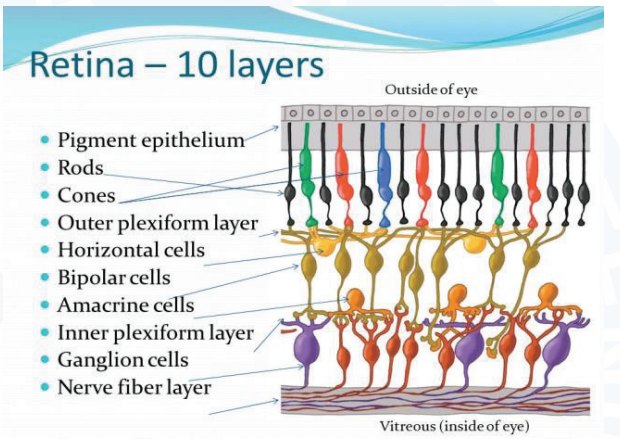
DIRECT OPHTHALMOSCOPY

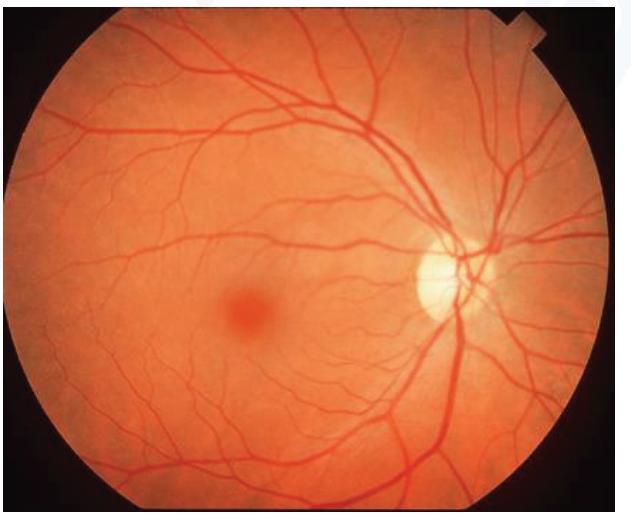
- Direct ophthalmoscope , close to face
- Virtual , erect , magnified image
- Magnification x 15
- To examine -,-,-

Active space

RETINA

- Innermost layer of eye - 10 Layers- 9 NSL /1 RPE
- Blood supply of retina
  - Inner 2/3: CRA
  - Outer 1/3: PCA , also choroid



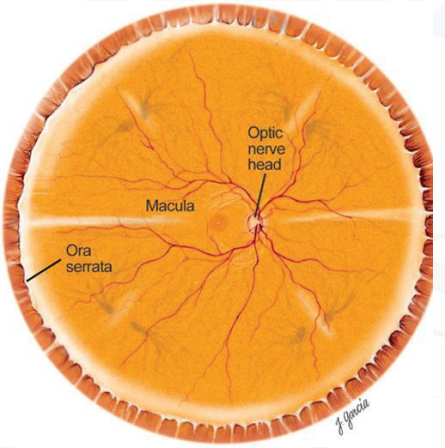


Active space

INDIRECT OPHTHALMOSCOPY

- Examines entire retina from fovea to ora serrata
- Drug of choice for dilating pupil –
- Real, inverted, magnified – x 5

  3 D picture of retina – seen binocularly



RETINAL DETACHMENT

- Separation of NSL from RPE
- Sudden painless loss of vision
- Floaters (Muscae)
- Flashes (Photopsiae)
- Curtain descending in front of eye
- O/ E - Normal red glow
- replaced by grey glow

Active space

TYPES OF RETINAL DETACHMENT

- Rhegmatogenous
- Tractional
- Exudative

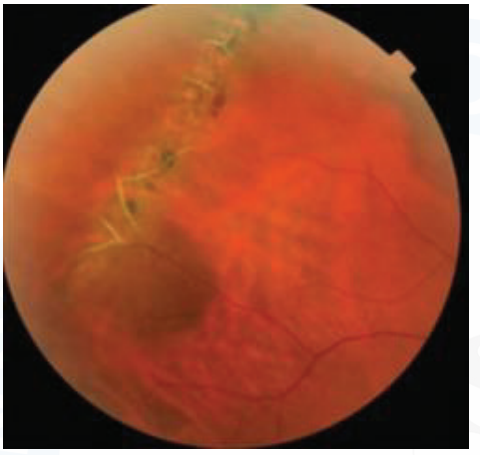
RISK FACORS RHEGMATOGENOUS RD

- High myopia (> 6 D)
- Post cataract surgery
- Trauma

Active space

LESIONS PREDISPOSING TO RD

- Holes
- Lattices
- Horseshoe tears



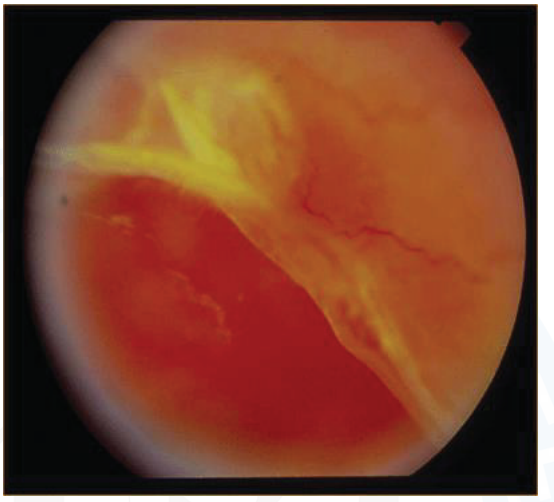
APHAKIA

- Vitreous supported by lens
- After cataract surgery , no support to vitreous , may cause detachment

Active space

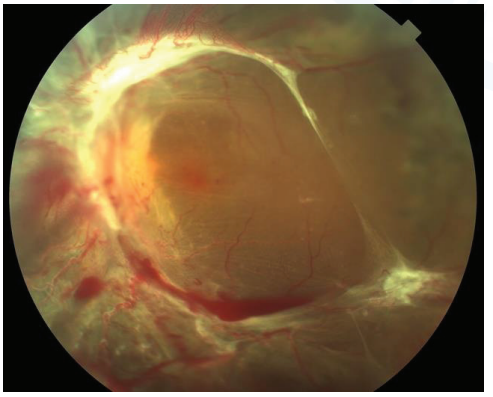
DIALYSIS

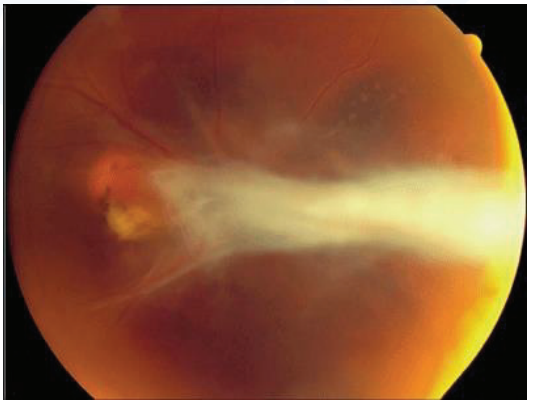
- Ocular trauma causes retinal dialysis -may cause a break allowing vitreous to cause RD



CAUSES OF TRACTIONAL RD

- Diabetic retinopathy
- Retinopathy of Prematurity
-

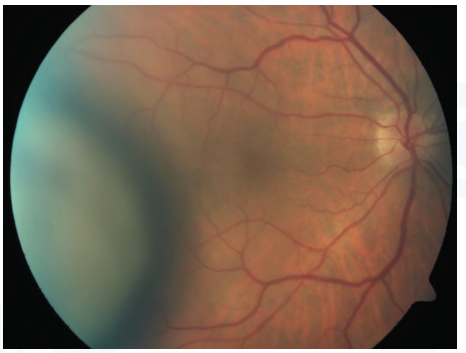


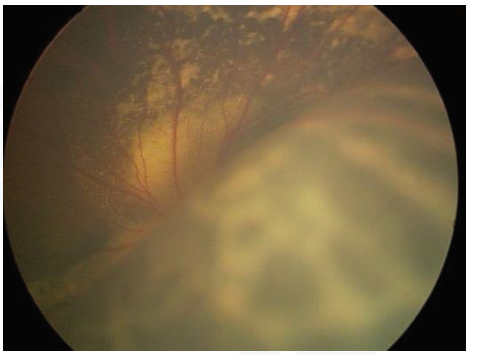


Active space

CAUSES OF EXUDATIVE RD

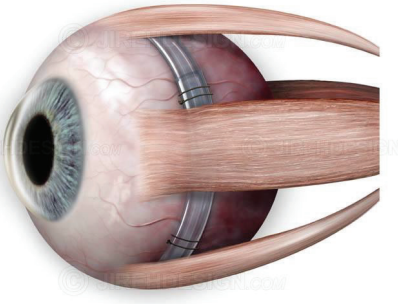
- Malignant melanoma of choroid
- Pregnancy induced hypertension





MANAGEMENT OF RD

- Reattach retina
- Rhegmatogenous – Buckling
- Exudative - treat underlying condition
- Tractional – Vitrectomy



Active space

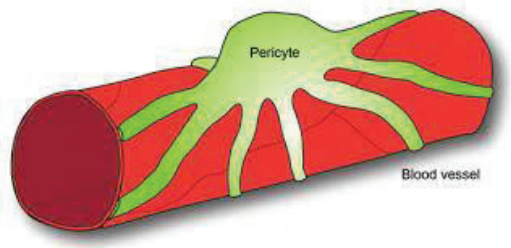
DIABETIC RETINOPATHY

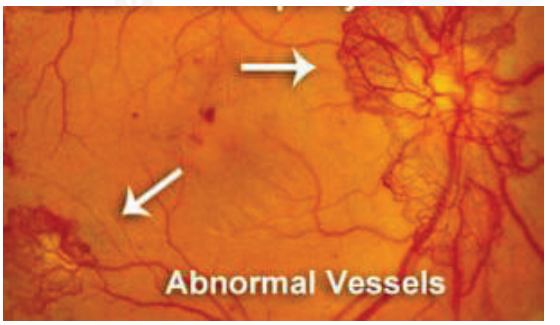
- WHO – Diabetic capital of the world
- Duration – Most important prognostic factor
- Control of blood glucose
- Hypertension /Hyperlipidemia /Pregnancy
- Type 1 /IDDM – 5 yrs
- Type 2 /NIDDM – at time of diagnosis
- Fluctuating vision – hyperglycemia –myopia

PATHOLOGY

Occlusion- hypoxia – VEGF -Neovascularization

- Pericyte necrosis – leakage

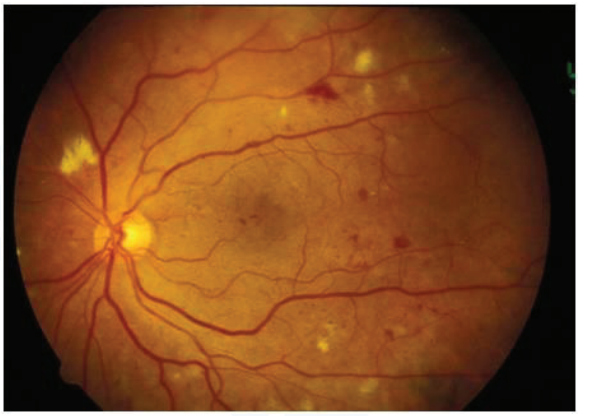




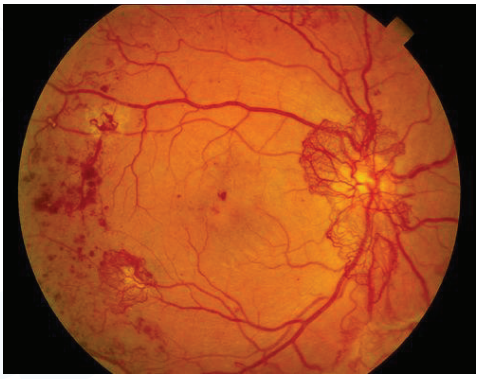
Active space

CLASSIFICATION

- Non -Proliferative Diabetic Retinopathy NPDR



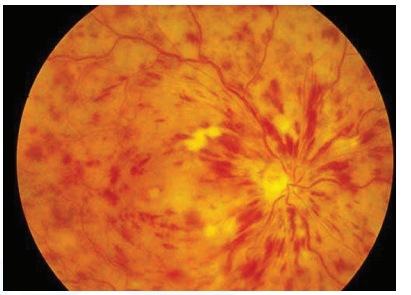
Proliferative Diabetic Retinopathy PDR

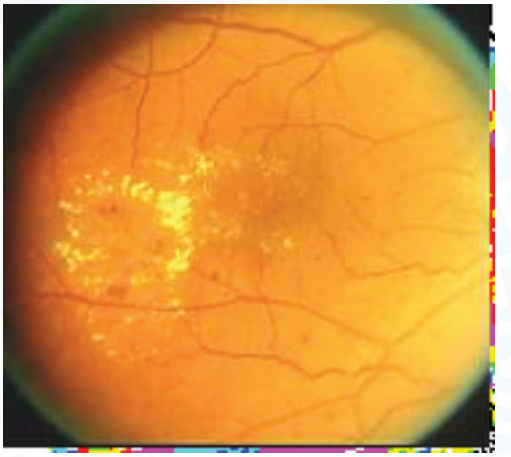


Active space

NPDR

- Earliest – Microaneurysms
- Dot and Blot haemorrhages
- Flame shaped haemorrhages
- Hard exudates
- Soft exudates





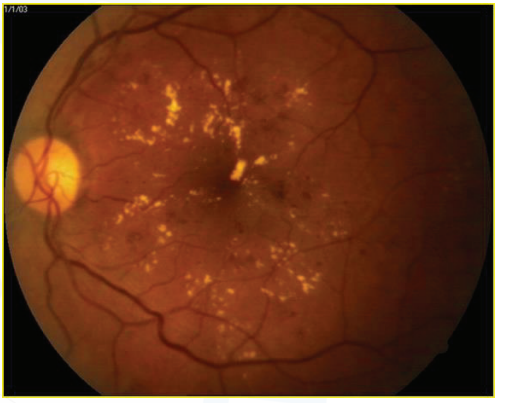
PDR

- Neovascularization of disc NVD
- Neovascularization elsewhere NVE
- Vitreous haemorrhage

Active space

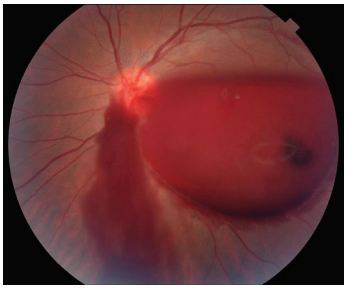
VISUAL LOSS IN DR

- NPDR
- Macular edema
- PDR
- Vitreous haemorrhage
- Tractional RD
- Neovascular glaucoma



VITREOUS HAEMORRHAGE

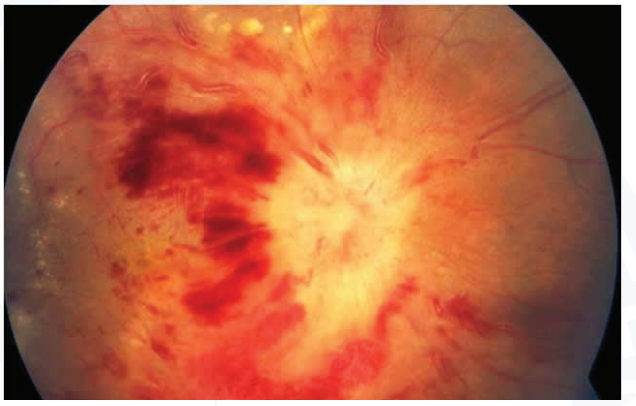
- Sudden painless loss of vision , red tint, floaters
- PDR – commonest
- PVD- posterior vitreous detachment
- Trauma – commonest in young
- Eales' disease



Active space

EALES' DISEASE

- Periphlebitis retinae
- TB
- Spontaneous, recurrent VH in young males
- ATT followed by steroids



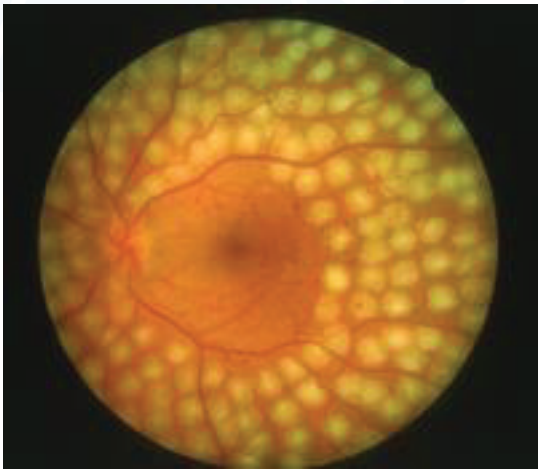
MANAGEMENT OF VITREOUS HAEMORRHAGE

- Bed rest x 3 months with head elevation
- Treat underlying condition- DM / ATT /Trauma
- Vitrectomy

Active space

ANTI VEGF DRUGS – NPDR

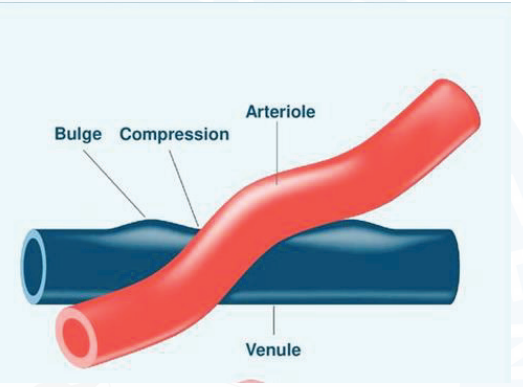
PRP FOR RETINAL HYPOXIA



Active space

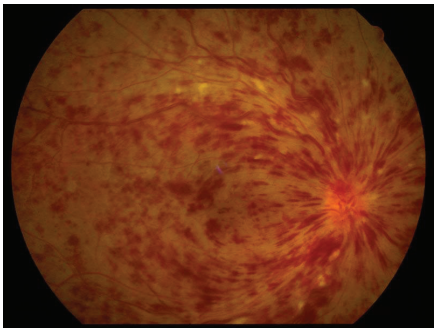
CENTRAL RETINAL VEIN OCCLUSION

- Risk factors
- Age > 65
- Hypertension
- Diabetes
- Hyperlipidemia
- Young ladies – OC



SIGNS/SYMPTOMS

- Sudden blurring of vision –like a voltage drop
- O/E – extensive haemorrhages on retina - blood and thunder fundus / tomato ketchup fundus
- " 90 day glaucoma " – NVG after 3 months CRVO
- m/c/c of loss of vision – macular oedema

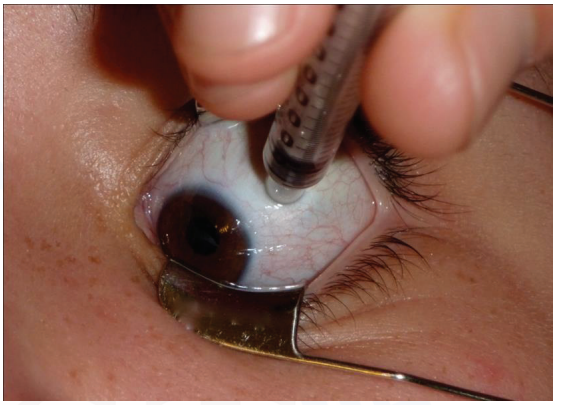


MIST

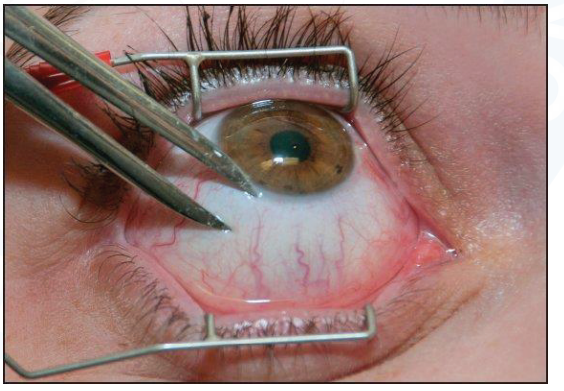
Active space

TREATMENT

- Anti -VEGF Drugs given intravitreal
- Bevacizumab /Ranibizumab



PARS PLANA SURFACE ANATOMY



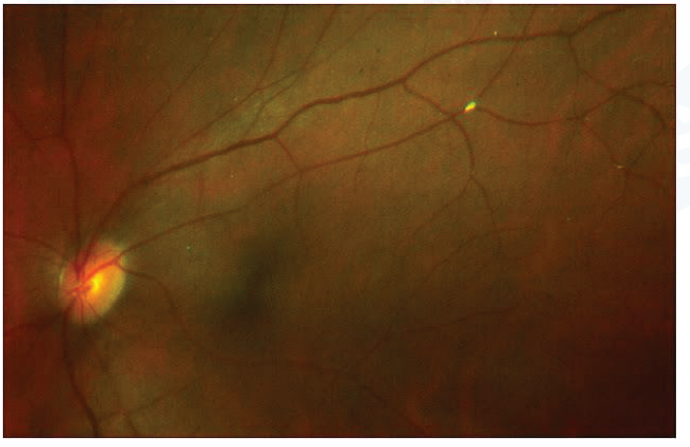
Active space

CENTRAL RETINAL ARTERY OCCLUSION

- Risk factors
- Advanced age
- Diabetes
- Hypertension
- Hyperlipidaemia
- Smoking

CRAO

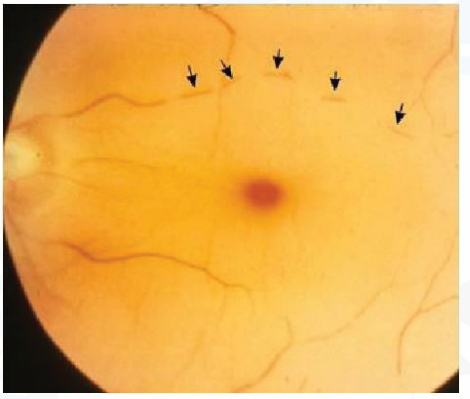
- Major ocular emergency -equivalent cerebral stroke
- Embolism – cholesterol m/c (Hollenhorst plaque)



Active space

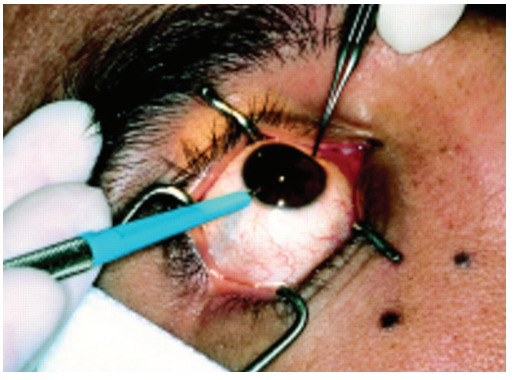
SIGNS/SYMPTOMS

- Sudden loss of vision
- Cherry red spot
- Cattle truck/ Box car appearance – segmentation of blood column
- Retinal Ischemic time – 90 minutes
- Partial recovery till 240 minutes



MANAGEMENT OF CRAO

- Irreversible after 4 hours
- Ocular massage
- Immediate lowering of IOP
- Paracentesis
- Carbogen
- I/V heparin, TPA

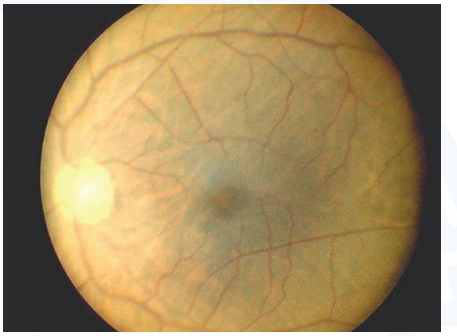


Active space

CYSTOID MACULAR EDEMA

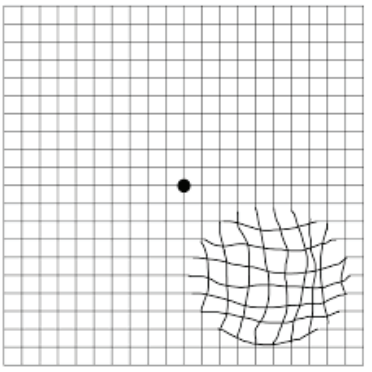
Causes

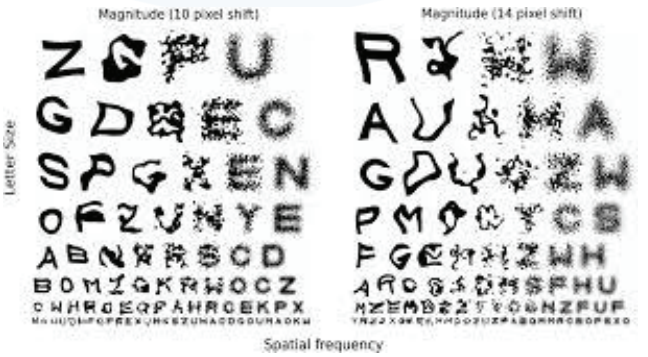
- D - Diabetes
- I - Irvine Gass syndrome
- Retinitis pigmentosa
- Vascular occlusions - CRVO



SIGNS/SYMPTOMS

- Painless blurring of vision
- Metamorphopsia
- Detected by Amsler's grid
- Fundus examination by 90 Dioptres lens /DO





Active space

+ 90 D FUNDUS EXAM

INVESTIGATIONS

- FFA –Fluorescein angiography
- OCT – optical coherence tomography

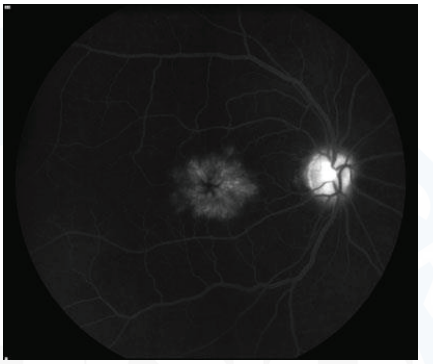
Active space

FLUORESCEIN ANGIOGRAPHY

Sodium Fluorescein dye injected antecubital vein

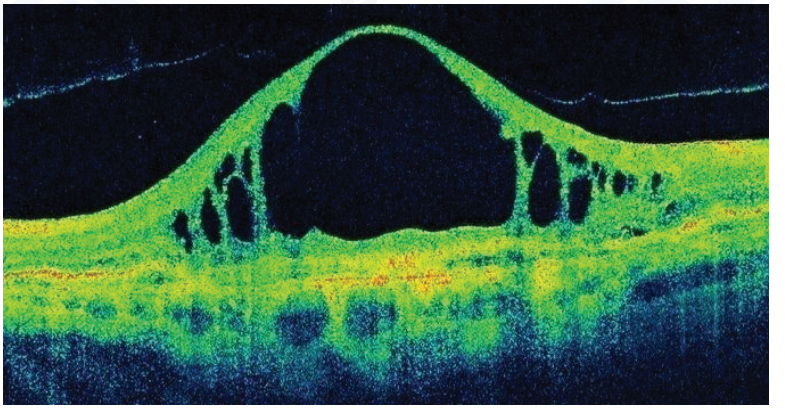
ART – arm retinal circulation time – 12 seconds

Flower petal/ Petaloid appearance – confirms CME



OPTICAL COHERENCE TOMOGRAPHY

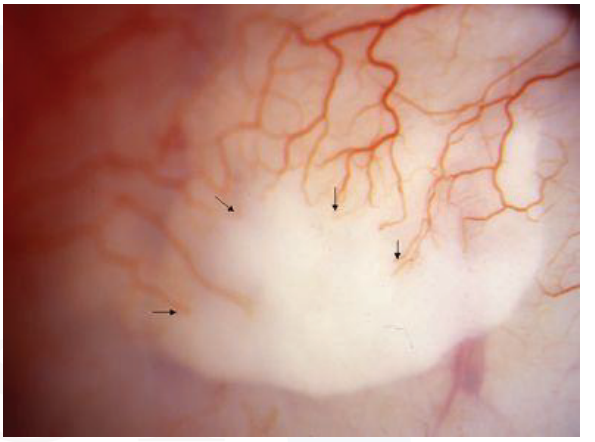
- Non invasive technique using interferometry to see cross section of retina



Active space

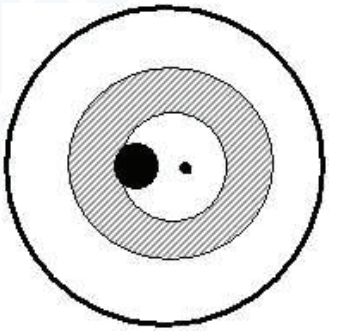
MANAGEMENT OF IRVINE GASS SYNDROME

- Topical steroids
- Topical NSAIDS
- Injection Triamcinolone sub Tenon's



RETINITIS PIGMENTOSA

- Apoptosis of rods
- AR/AD/X-linked
- Risk factor -Consanguinity
- Nyctalopia- earliest
- Ring scotoma
- Constriction of visual fields
- End stage tunnel vision



Active space

- Triad
  - Pale waxy disc
  - Arteriolar attenuation
  - Bone spicule pigmentation

- Gold standard flat ERG

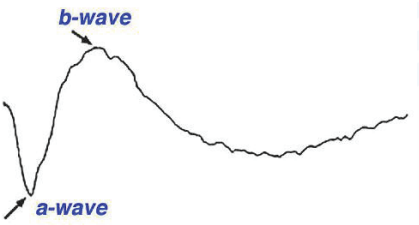
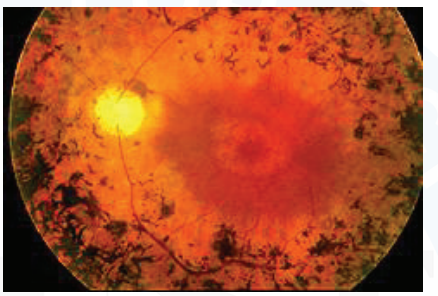


Fig.1 The biphasic waveform of the ERG of a normal patient.



Treatment:

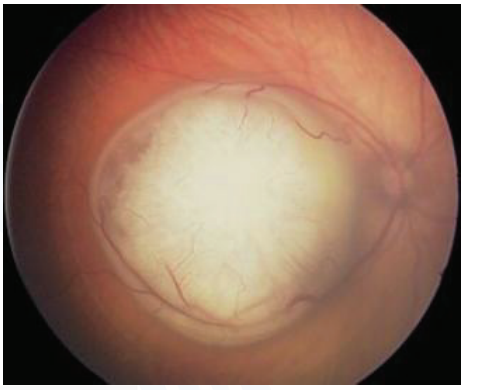
- 15,000 IU/day of Vit A in Palmitate form with 12 mg lutein every day
- Avoid Vitamin E
- Usually become blind by 40 but 7 years of vision can be retained by Vit A

BIONIC EYE / ARGUS II

Active space

RETINOBLASTOMA

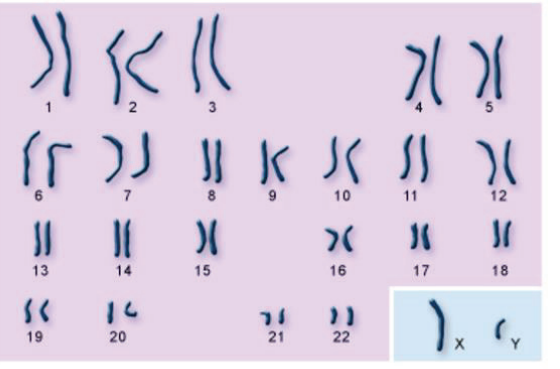
- Commonest ocular tumor of childhood
- Bilateral 30%
- Unilateral 70%



- RB gene (RB1)-suppressor gene – prevents tumors
- RB1 deleted /damaged

RB1 located on 13 q14

First human cancer suppressor gene to be located

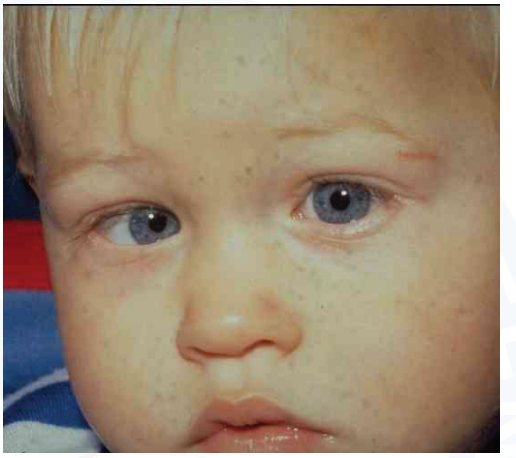


U.S. National Library of Medicine

Active space

Clinical Features

- Leucocoria (Amaurotic cat's eye pupil)
- Strabismus
- Glaucoma –NVG
- Loss of vision



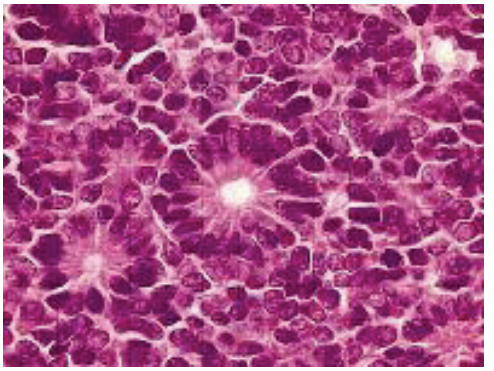
Active space

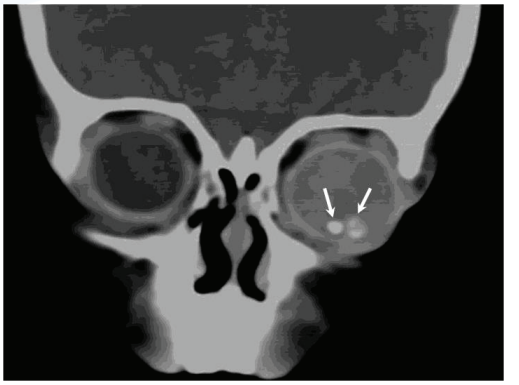
D/D OF LEUCOCORIA

- Retinoblastoma
- Coats Disease
- Congenital cataract

RETINOBLASTOMA

- Hallmark – Flexner-Wintersteiner Rosettes
- Commonest cause of intraocular calcification- MRI best
- Management
  - Primary: Save life
  - Secondary: Save eye
  - Tertiary: Save vision





Active space

INTERNATIONAL CLASSIFICATION OF RETINOBLASTOMA

A. ≤3 mm

B. >3 mm

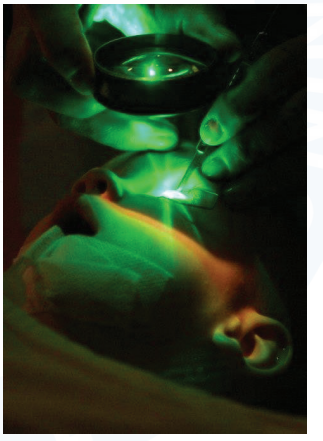
C. Localized seeds (< 3 mm from tumour)

D. Diffuse seeds (> 3 mm)

E. Massive Retinoblastoma > than 50%

MANAGEMENT OF RETINOBLASTOMA

- Stage A
  - Cryotherapy
  - Laser photocoagulation



INTRAOCULAR

Stage B/C/D

Chemotherapy – Intravenous , intra vitreal , intra arterial

M/c regimen - VEC – 6 cycles

- Vincristine Etoposide Carboplatin

Intra arterial – Ophthalmic artery

Active space

STAGE E -ENUCLEATION

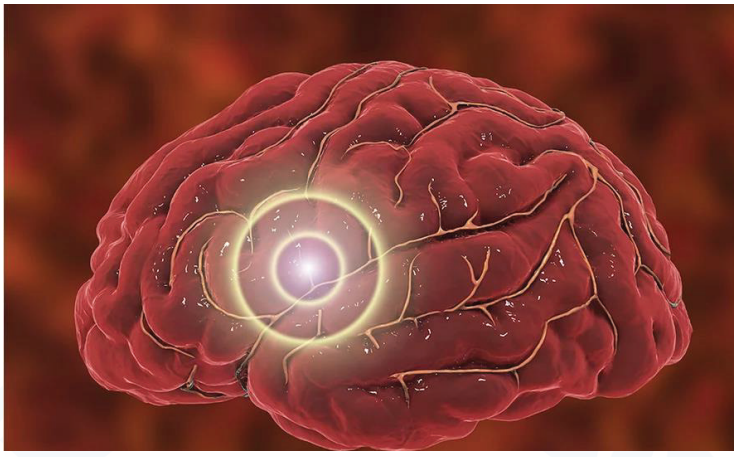
- Delaying enucleation may be risky
- Sacrifice 15 mm of optic nerve
- Hydoxyapatite orbital implant

CAUSES OF DEATH

- Metastases - m/c optic nerve
- Intracranial tumours associated with RB
- Pinealoblastoma with bilateral retinoblastoma –
- --
- M/c Secondary tumour associated with RB –

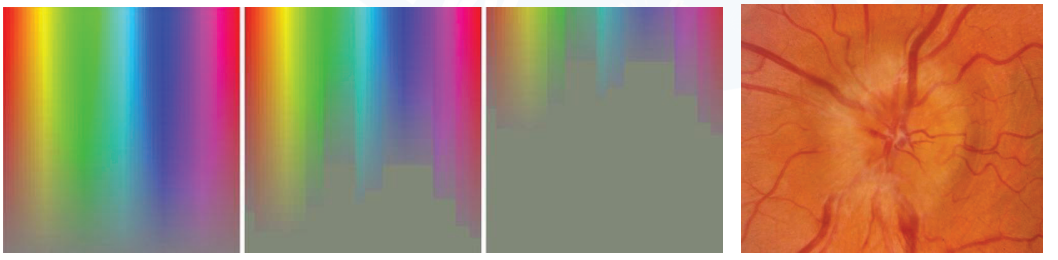
Active space

NEUROOPHTHALMOLOGY



OPTIC NEURITIS/ RETROBULBAR NEURITIS

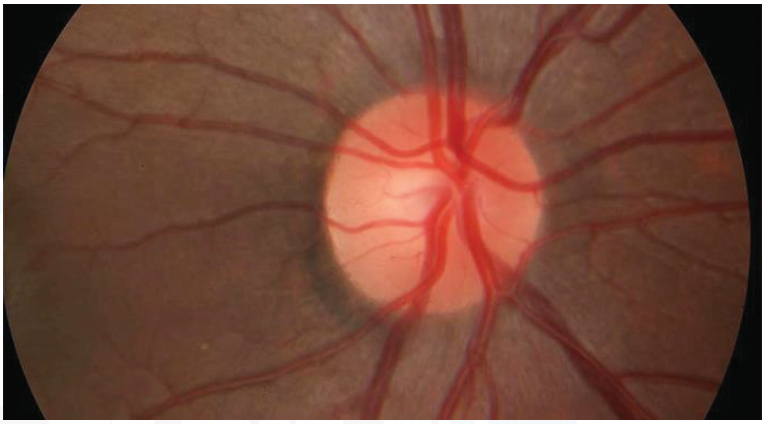
- M/c/c multiple sclerosis –young ladies
- Sudden painful loss of vision
- Pain with ocular movements
- Color vision desaturation
- Abnormal pupillary reaction-RAPD/Marcus Gunn Pupil
- Hallmark: Disc oedema



Active space

RETROBULBAR NEURITIS

- Absence of disc oedema
- " Condition where patient sees nothing and doctor sees nothing "



MANAGEMENT

- Visual field defects – Central /Centrocecal scotoma
- IVMP –Intra venous methyl prednisolone
- Oral steroids contraindicated

Active space

IMPORTANT NO VISUAL FIELD DEFECTS

Central/ Centrocecal scotoma

Enlargement of blind spot

Optic neuritis

Papilledema

FIELD DEFECTS

- Altitudinal hemianopia -AION (Anterior ischemic optic neuropathy)

  Central/Centrocecal scotoma Toxic /Nutrition ON

  Tobacco

  Alcohol

  Ethambutol

  Tamoxifen

  Amiodarone

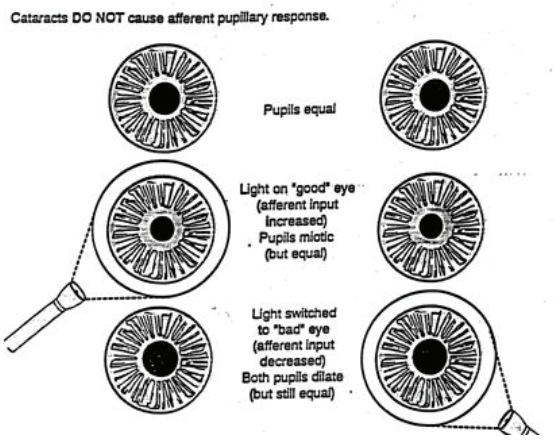
Active space

PUPILLARY LIGHT REFLEX

- Direct – pupil under torchlight constricts on exposure to light
- Consensual reaction – opposite eye reaction

MARCUS GUNN PUPIL/RAPD

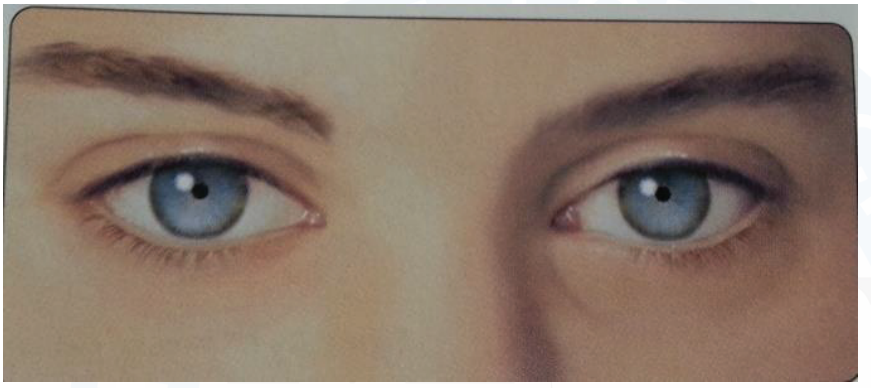
- Abnormal pupillary reaction seen in
- Unilateral/Asymmetric optic nerve damage- optic neuritis / retrobulbar neuritis / AION
- Pupil dilates to light rather than constricting



Active space

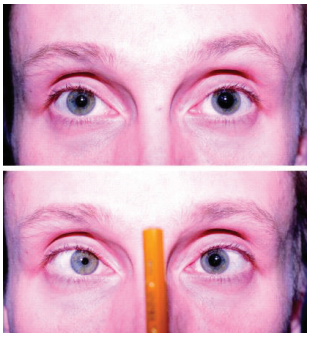
ARGYLL ROBERTSON PUPIL

- Seen in Neurosyphilis / tertiary syphilis
- Bilateral , constricted , irregular pupil
- Light negative , accommodation positive
- ' Prostitute's pupil '



ADIE'S PUPIL

- Seen in young ladies with a viral fever
- One pupil dilates , the other remains normal- due to damage of parasympathetic fibres of 3rd CN
- Light Near dissociation – does not constrict to light but constricts to accommodation
- Similar to Argyll Robertson – light reaction negative , accommodation positive



Active space

CONFIRMATORY TEST FOR ADIE'S

- Pilocarpine 0.125 % test – normal pupil does not constrict , Adie's pupil does
- Denervation supersensitivity

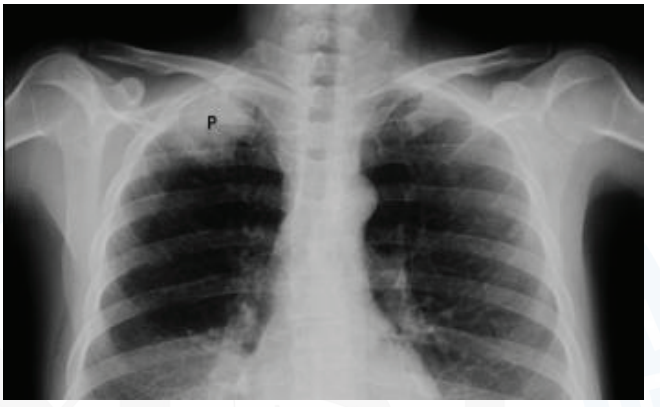
HORNER'S SYNDROME

- Oculo sympathetic paralysis
- Ptosis /Miosis / -
- Congenital and Acquired
- Congenital has -

Active space

ACQUIRED HORNER'S SYNDROME

- Causes
- Pancoast tumour
- Internal Carotid artery rupture

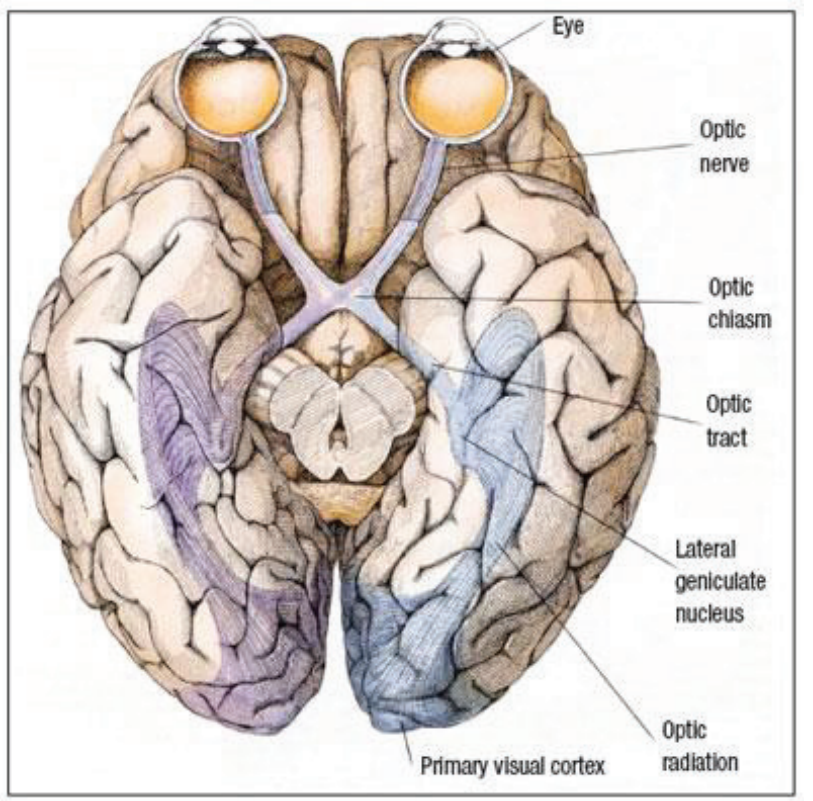


HORNER'S SYNDROME

- Congenital Horner's – birth trauma due to forceps delivery
- Cocaine Test for confirmation – normal pupil dilates , HS pupil doesn't

Active space

VISUAL PATHWAYS

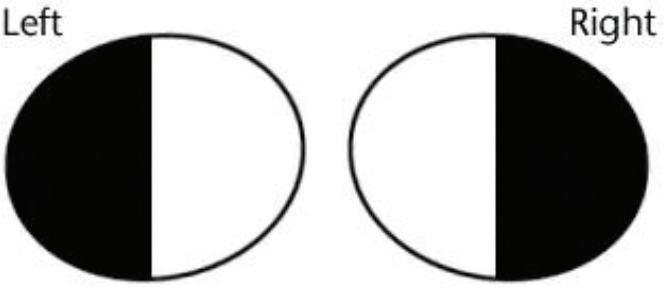


VISUAL FIELD DEFECTS

Active space

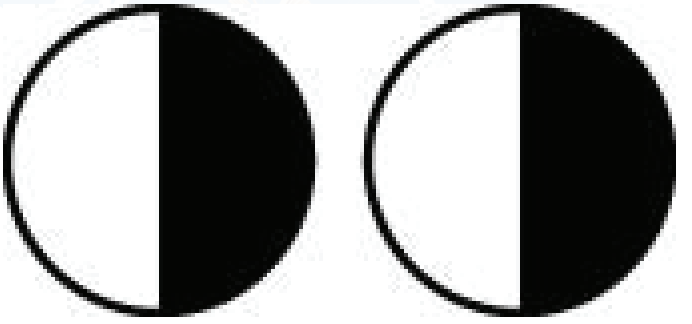
BITEMPORAL HEMIANOPIA

- Bitemporal hemianopia – loss of both temporal halves of the visual field
- Optic chiasmal lesion
- m/c – Pituitary adenoma
  Craniopharyngioma



HOMONYMOUS HEMIANOPIA

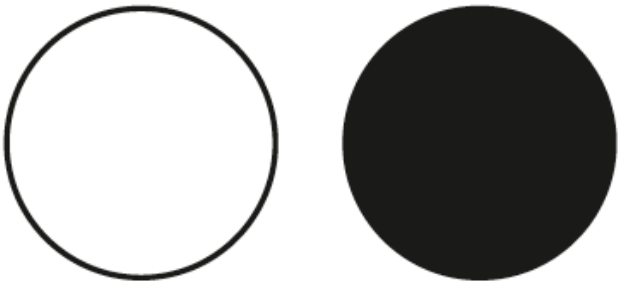
- Visual field loss in the same half of the fields in each eye
- Can be caused by any contralateral retro chiasmatic lesion – optic tract ,LGB , optic radiations , occipital lobe
- M/c/c occipital lobe lesions



Active space

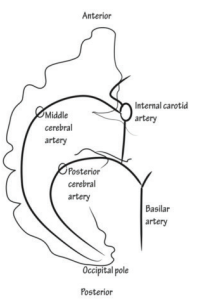
MONOCULAR FIELD LOSS

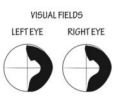
- Pre chiasmal lesion
- Same sided lesion



HOMONYMOUS HEMIANOPIA WITH MACULAR SPARING

- Occipital lobe
- Dual blood supply – PCA and MCA





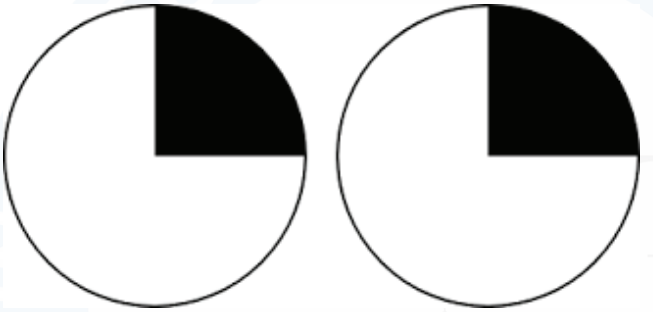
Active space

SUMMARY VISUAL FIELD DEFECTS

Optic nerve	Monocular blindness
Optic chiasma	Bitemporal hemianopia
Optic tract	Homonymous hemianopia
LGB	Rarely seen

Optic radiations - Meyer's loop – " Pie in the sky"

Occipital lobe	Macular sparing



STRABISMUS / SQUINT

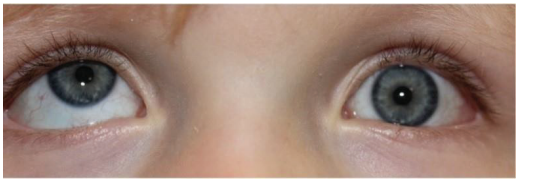
Active space

DEFINITIONS

- Orthophoria – eyes are parallel
- Heterophoria – eyes are NOT parallel (strabismus)
- Tropia – manifest squint
- Phoria - Latent squint – held by binocular fusion
- Manifests when eye muscles tired

TYPES OF STRABISMUS

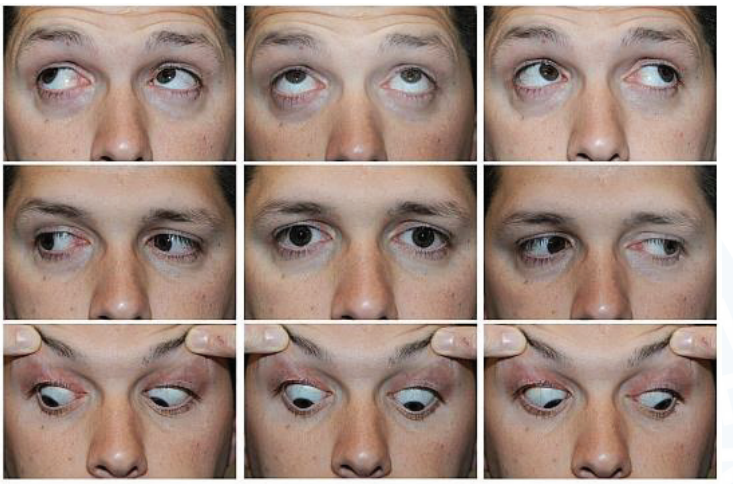
- Exotropia – one eye moves out
- Esotropia – One eye moves in
- Hypertropia – One eye is above the other
- Hypotropia – one eye is below the other



Active space

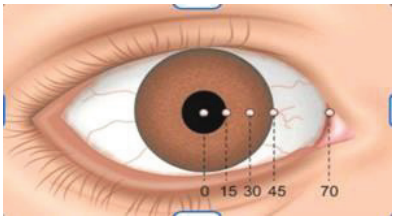
9 GAZES

- 1 primary gaze - head straight , eyes straight
- 4 secondary gazes
- 4 tertiary gazes



TEST FOR DETECTION OF TROPIAS

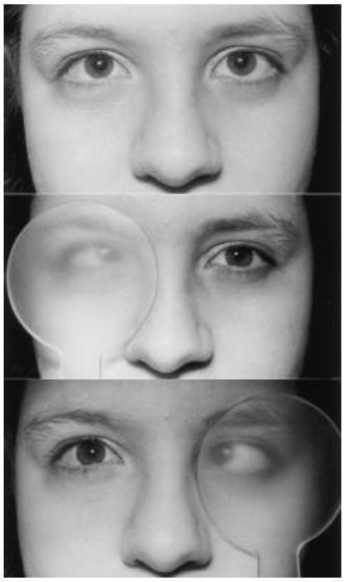
- Hirschberg test –
- Each 1 mm of decentration of light reflex = 7 degrees of squint
- 1 degree = 2 prism dioptres PD
- Light reflex on pupil margin= 15 degrees
- Light reflex on limbus = 45 degrees



Active space

TEST FOR DETECTION OF PHORIAS

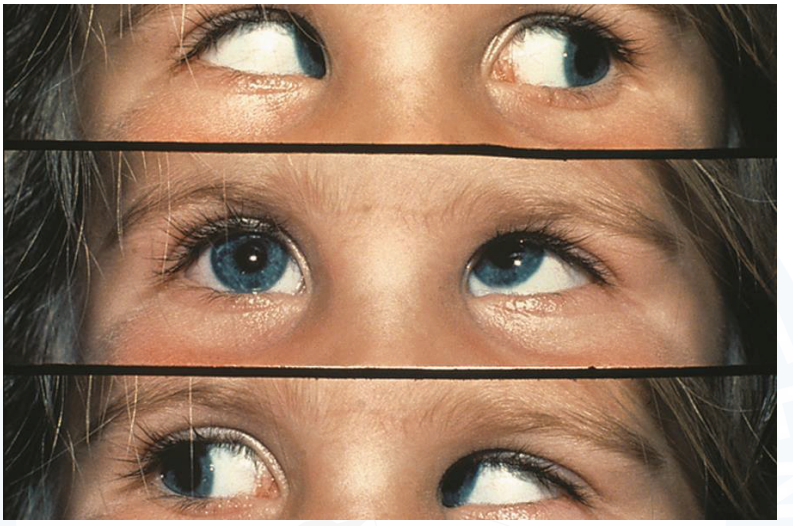
- Cover uncover test – cover one eye and uncover it
- Eye does not move – orthophoria
- Eye moves from in to out – esophoria
- Eye moves from out to in - exophoria



Active space

TYPES OF TROPIAS

- Comitant - type of squint where the angle between the eyes remain same in all 9 gazes
- No functional problem



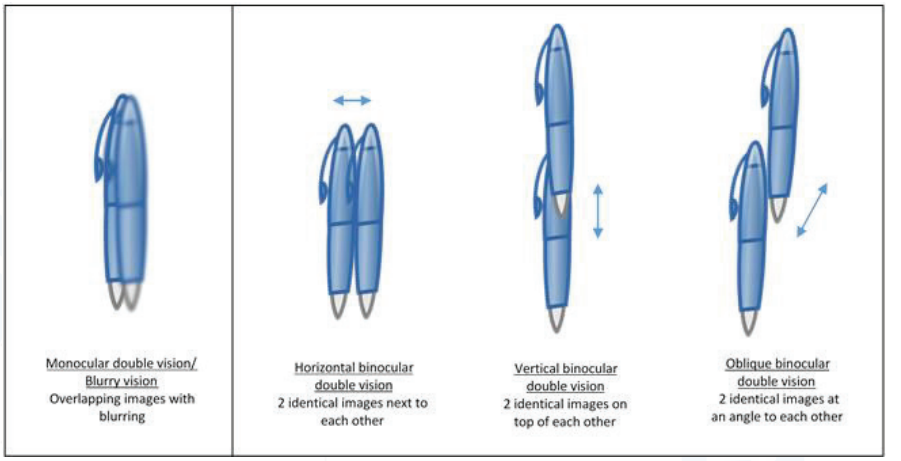
TYPES OF SQUINT

- Paralytic squint –
- Due to paralysis of nerves(3, 4 ,6th CN) or NMJ
- Causes – vasculopathic causes – DM , Hypertension
- Diplopia
- Abnormal head posture
- Vertigo
- Disorientation

Active space

CLASSIFICATION OF DIPLOPIA

- Horizontal - Either LR /MR palsy
- Vertical - Either SR/IR palsy
- Oblique - Either SO /IO palsy



MONOCULAR VS BINOCULAR DIPLOPIA

Active space

MONOCULAR / BINOCULAR DIPLOPIA

- Monocular
- Cataract
- Keratoconus
- Astigmatism
- Dry eyes

Binocular

- Strabismus

ABNORMAL HEAD POSTURE-FACE TURN

- Interpretation – horizontal muscle palsy –LR/MR

Active space

AHP – CHIN LIFT / DEPRESSION

- Interpretation – vertical muscle palsy –SR/IR

AHP – HEAD TILT

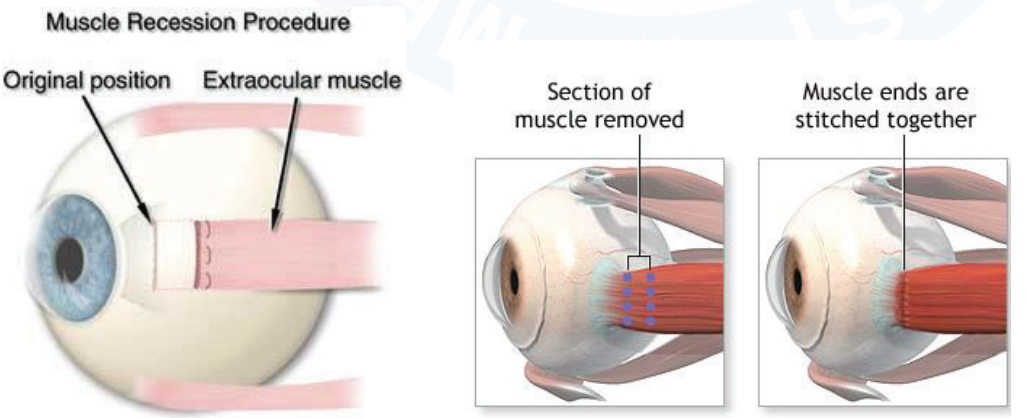
- Interpretation – oblique muscle palsy – SO/IO

Active space

TYPES OF COMITANT SQUINT

- Accommodative – due to uncorrected refractive error
- Correct refractive error-prescribe glasses

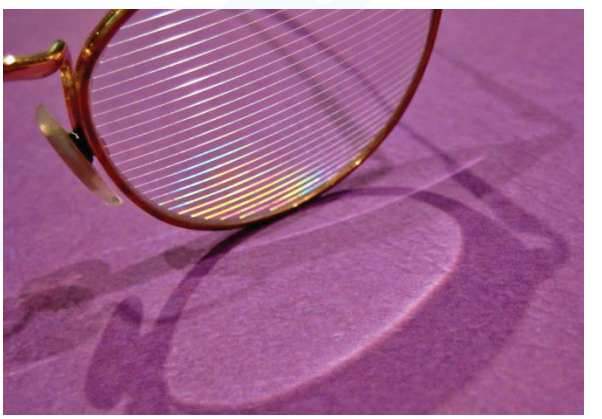
- Non accommodative squint
- NOT due to refractive error
- Surgical correction –
- Weaken one muscle – recession
- Strengthen opposite muscle –resection



Active space

MANAGEMENT OF PARALYTIC SQUINT

- Control underlying disorder – treat DM / hypertension
- Wait and watch for 3-6 months for nerve /muscle recovery
- Control the diplopia – Patching
- Prismatic glasses
- Botulinum toxin



Active space

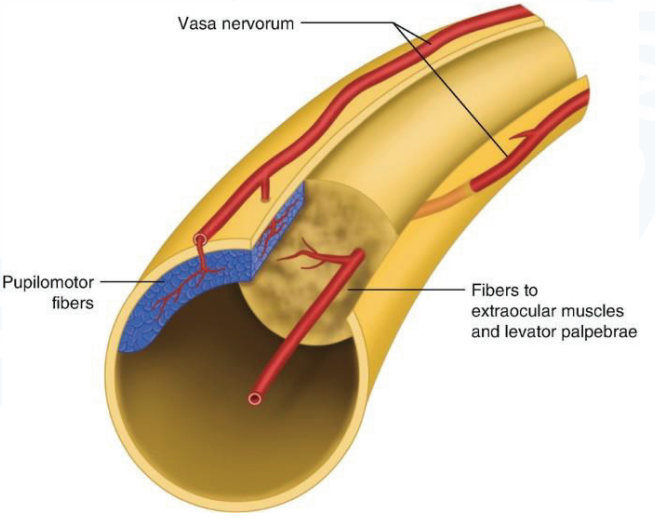
3RD CN PALSY

Diplopia – presenting complaint

"Down and out " with ptosis

Outer parasympathetic fibres supplying Sphincter pupillae and ciliary muscle

Inner somatic fibres supply SR ,IR ,MR , IO , LPS



Active space

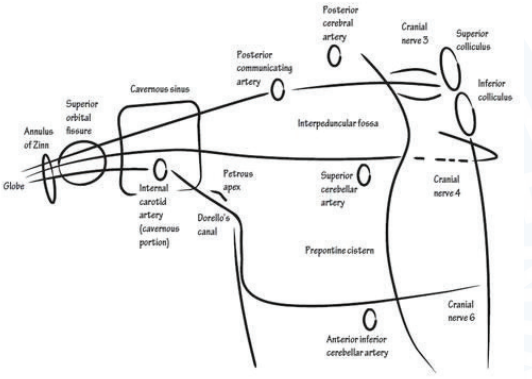
3RD CN PALSY

- Causes
- Vascular ischemia – DM / Hypertension- pupil sparing – normal pupil- 'Medical' 3rd nerve
- Tumours / Aneurysms –Pupil involving – dilated pupil – ' Surgical ' 3rd nerve

Active space

4TH CN PALSY

- Longest intracranial nerve
- Least number of axons
- - cranial nerve
- Most common nerve to be involved in -
- Only nerve to --
- Only nerve to exit dorsally from brainstem



4TH CN PALSY

- Causes
- Traumatic , congenital , vasculopathic

Vertical diplopia on looking downwards/ inwards – reading

Position of eyes – upwards , head tilt on opposite shoulder

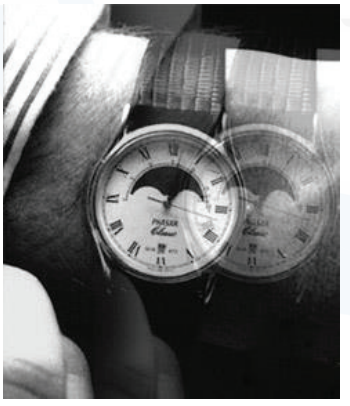
How text appears if you have double vision

How text appears if you have double vision

Active space

6TH NERVE PALSY

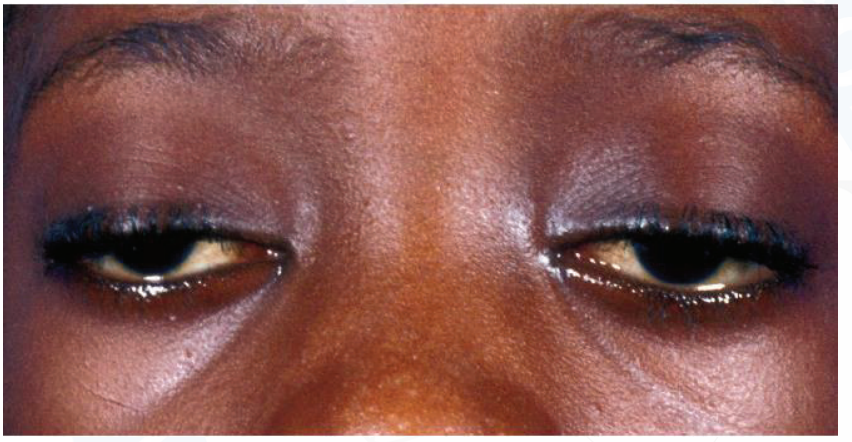
- Longest sub arachnoid nerve/ second longest intracranial
- M/c ocular motor palsy
- M/c to be affected in lumbar puncture
- Causes - Vasculopathies , children – neoplasms
- Horizontal diplopia , esotropia, face turn out



Active space

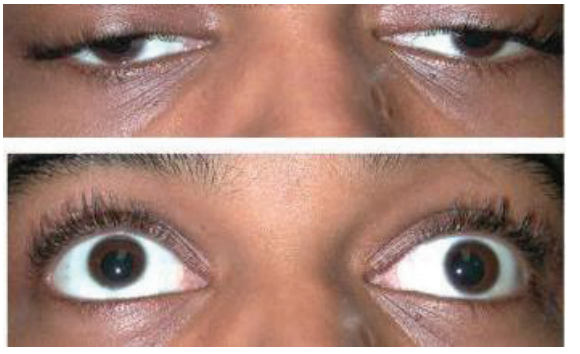
MYASTHENIA GRAVIS

- M/c disorder of neuromuscular junction
- Autoimmune disorder→ antibodies against Acetylcholine receptors Ach R at NMJ → impaired transmission → Muscle weakness particularly with fatigue-myasthenic fatigue
- Women – 20- 40years / Men- 60 to 80 years
- Fluctuating ptosis and diplopia worsening evening



MYASTHENIA GRAVIS

- Diagnostic tests – Tensilon test –Injection edrophonium reverses MG immediately



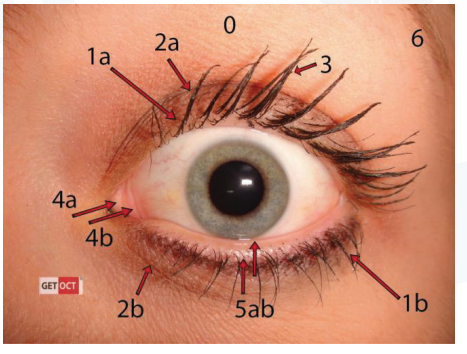
Active space

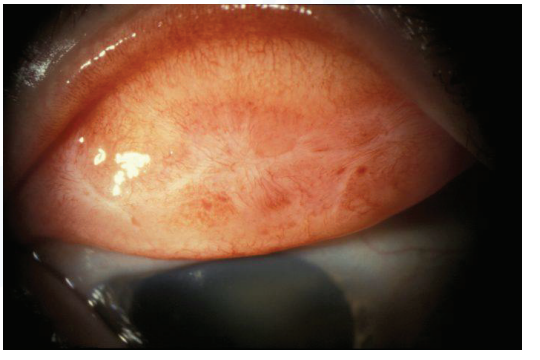
MIST

- Icepack test -placing icepack on closed lids reverses ptosis

CONJUNCTIVA

- Transparent membrane covering eyeball/ eyelids
- Functions - protection/ lubrication /immune surveillance
- Has goblet cells producing mucin

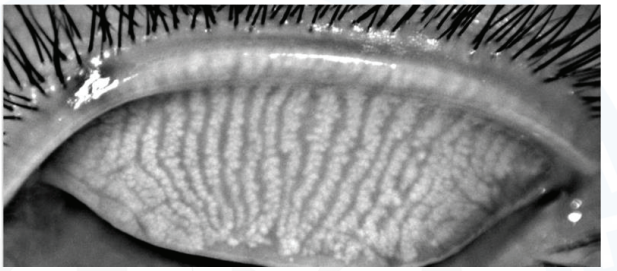


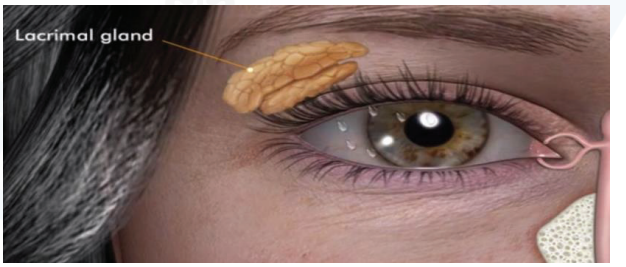


Active space

STRUCTURE OF TEAR FILM

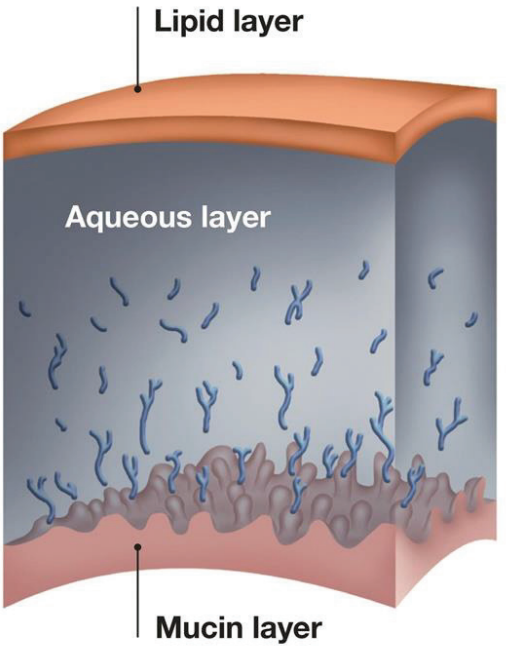
- Outer lipid layer – Meibum – secreted by Meibomian glands -prevents evaporation
- Middle aqueous layer – secreted by lacrimal glands lubrication / protection/ nutrition to cornea
- Inner mucin layer – goblet cells – stabilize tear film
- Deficiency of any-Dry Eyes/ Keratoconjunctivitis Sicca



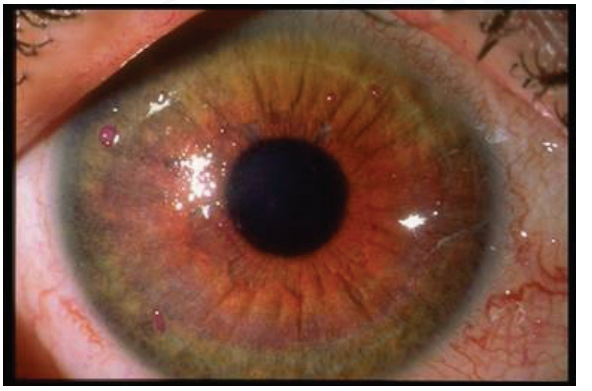


Active space

TEAR FILM STRUCTURE



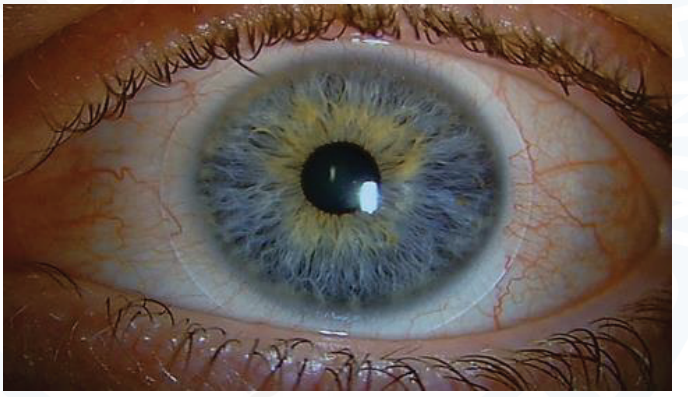
DRY EYES



Active space

CAUSES OF DRY EYES

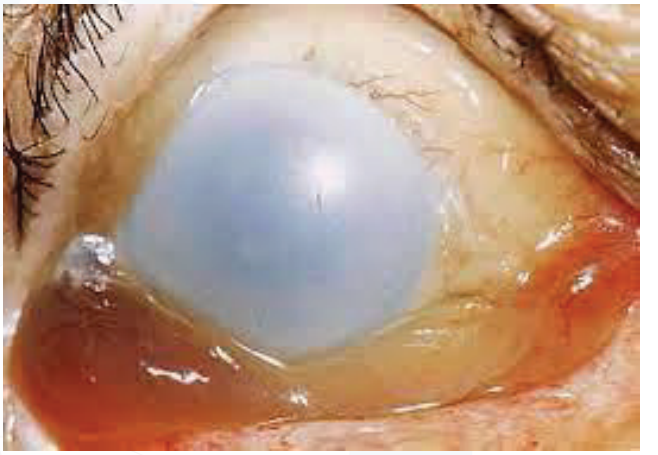
- Lipid deficiency –MGD
- Aqueous deficiency
- Old Age
- Prolonged contact lens wear
- -
- Sjogren's syndrome – Dry eyes / dry mouth /RA
- Drug induced – anti cholinergics
  anti hypertensives
- -



Active space

CAUSES OF DRY EYES

- Mucin deficiency
- Vitamin A deficiency
- Trachoma
- Chemical injury



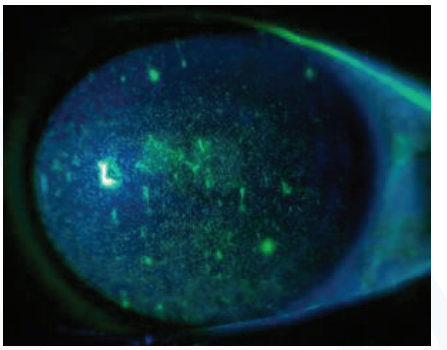
SYMPTOMS OF DRY EYES

- Foreign body sensation – burning / itching /gritty
- Redness of eyes
- Blurring of vision
- Intolerance to fans /AC's
- Worsening of symptoms in the evening
- Light sensitivity
- Pain while wearing contact lenses
- Watering of eyes

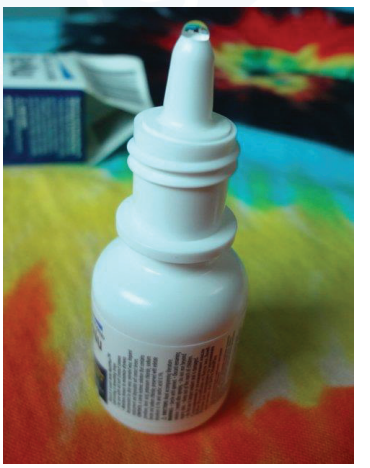
Active space

INVESTIGATIONS FOR KCS

- Conjunctival dryness – Rose Bengal/Lissamine green
- Corneal dryness – Fluorescein stain
- Schirmer's test - < 10 mm in 5 minutes



TREATMENT -METHYL CELLULOSE

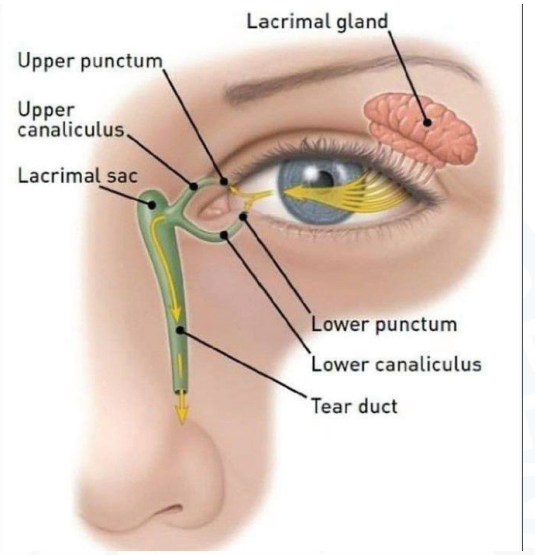


Active space

MIST

LACRIMAL DRAINAGE PATHWAY

- Lacrimal gland – lacrimal puncta – lacrimal canaliculi – common canaliculi – lacrimal sac – nasolacrimal duct – inferior meatus of nose



CONGENITAL NASOLACRIMAL DUCT OBSTRUCTION(NLDO)

- 5% children- obstruction of distal end of NLD – Hasner's valve not canalised
- Infection of stagnant tears in sac
- Epiphora, discharge, crusting
- D/ D - Congenital glaucoma- photophobia/blepharospasm

Active space

TREATMENT

- Lacrimal sac massage =(Crigler massage) upto 6 months
- Probing
- Dacryocystorhinostomy(DCR) – connect sac to the middle meatus of nose
- After 3 years of age

Active space

VIT A DEFICIENCY CLASSIFICATION

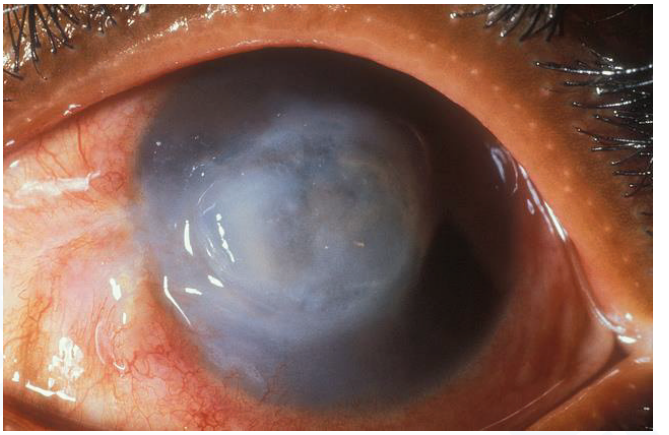
TABLE 2. WORLD HEALTH ORGANIZATION RE-CLASSIFICATION OF XEROPHTHALMIA SIGNS

Classification	Ocular Signs
XN	Night blindness
X1A	Conjunctival xerosis
X1B	Bitot's spots
X2	Corneal xerosis
X3A	Corneal ulceration-keratomalacia involving one-third or less of the cornea
X3B	Corneal ulceration-keratomalacia involving one-half or more of the cornea
XS	Corneal scar
XF	Xerophthalmic fundus

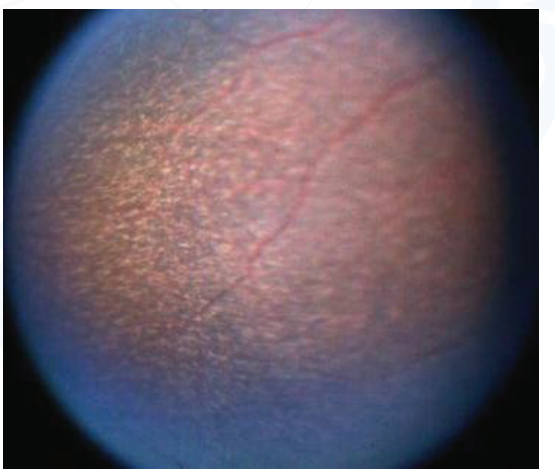
X1 B BITOT'S SPOT

Active space

KERATOMALACIA



XEROPHTHALMIC FUNDUS



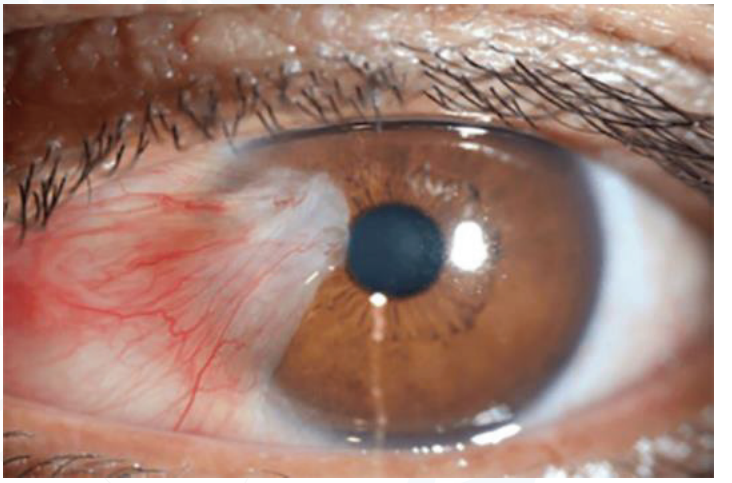
Active space

PTERYGIUM

Fibrovascular overgrowth of conjunctiva over the cornea

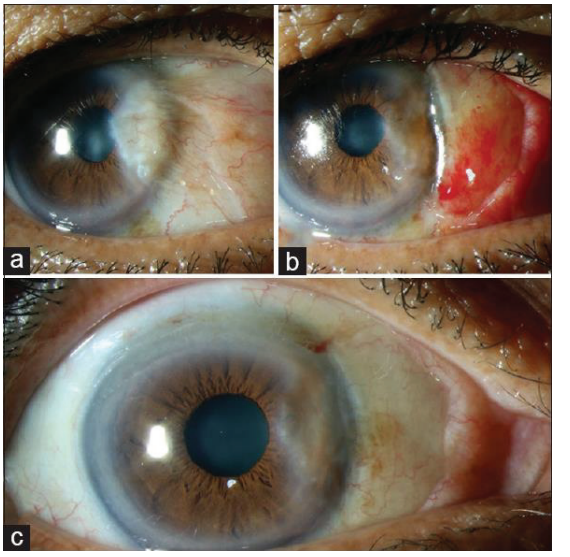
Redness , irritation , FB sensation, watering , photophobia inability to wear contact lenses

- Blurring of vision- astigmatism , blocks pupil
- Stocker's line at advancing head



MANAGEMENT

- Excision with conjunctival autograft



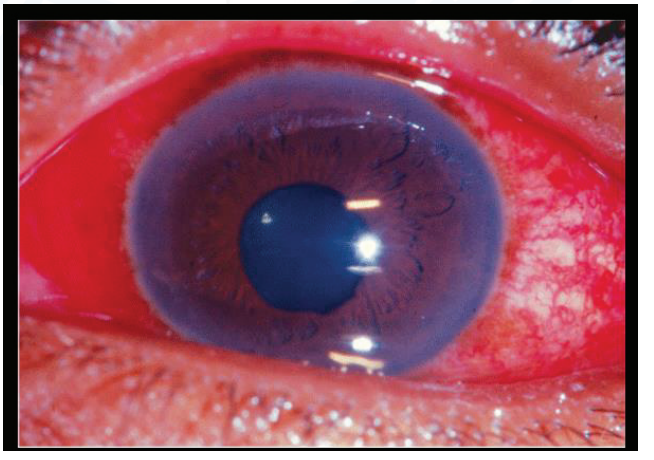
Active space

CONJUNCTIVITIS

- Infection of conjunctiva – infectious/non infectious
- M/c infectious cause – viral – Adenovirus
- Bright red in colour
- Painless
- Discharge –may be purulent , serous , mucoid
- Purulent – Bacterial
- Serous – Viral
- Mucoid – allergic

EPIDEMIC KERATOCONJUNCTIVTIS

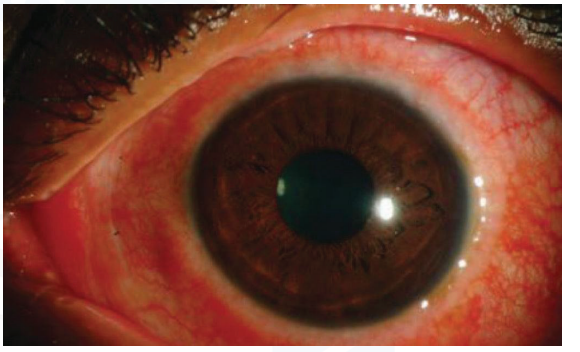
- M/c/c of conjunctivitis
- Adenovirus -Pink Eye /Madras Eye
- Same community
- Periauricular lymphadenopathy
- Symptomatic treatment



Active space

ACUTE HEMORRHAGIC CONJUNCTIVITIS

- EV – 70 , enterovirus
- Young adults
- Subconjunctival haemorrhage
- Apollo disease
- Cold compresses / astringents



TRACHOMA- KERATOCONJUNCTIVITIS

- "Forgotten disease of forgotten people "
- Egyptian Ophthalmia
- Chlamydia trachomatis – A, B, Ba ,C
- Commonest infective cause of blindness- corneal scarring

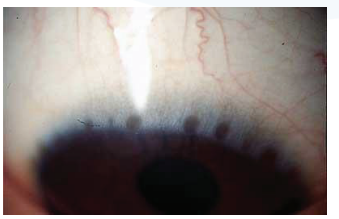
Active space

5 F – TRANSMISSION FACTORS

- Flies
- Fingers
- Fomites
- Faeces
- Faces

TRACHOMA

- Lacrimation, congestion ,discharge
- Children affected most , followed by women
- Sago grain follicles
- Herbert's pits
- Arlt's line

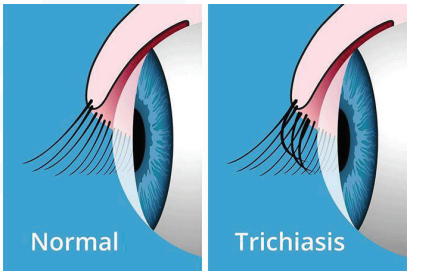


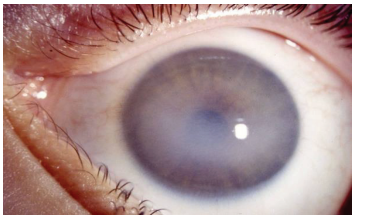
Active space

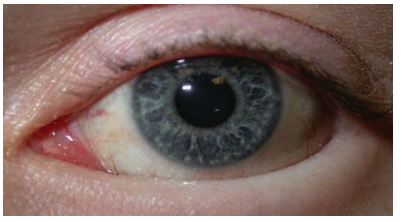
- SAFE strategy
- Surgery –for trichiasis
- Antibiotics -Azithromycin
- Facial cleanliness
- Environmental improvement
- Blanket therapy-
- 1% Tetracycline ointment once a day for 10 days in a month for 6 months

TRACHOMA SEQUELAE

- Trichiasis
- Tylosis
- Madarosis
- Dry eyes
- Entropion
- Corneal opacities
  - Nebula
  - Macula
  - Leucoma



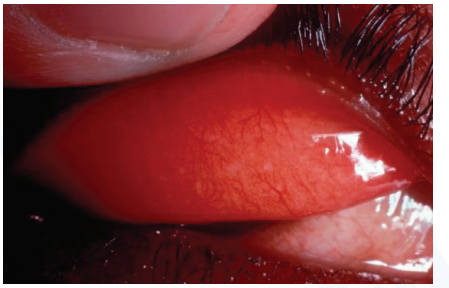




Active space

WHO GRADING

- F – Follicles > 5 , 0.5 mm or larger in diameter
- II. I - Inflammation – 50% tarsal vessels obscured
- III. S- Scarring - Arlt's line
- IV. T – Trichiasis
- V. O – Corneal opacity



VISION 2020

- Cataract
- Trachoma (SAFE strategy)
- Childhood blindness
- Refractive error
- Onchocerciasis- river blindness

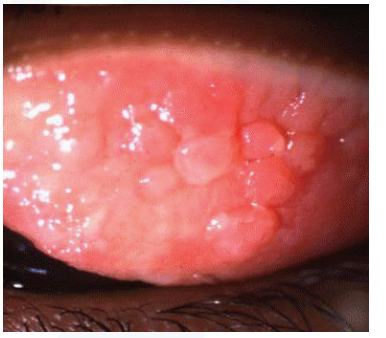
Active space

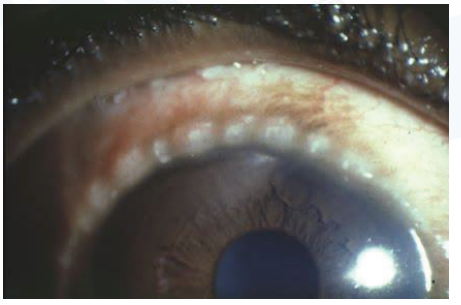
SPRING/ VERNAL CATARRH

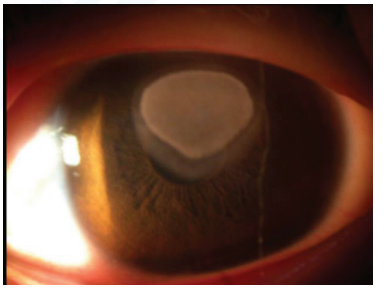
- Allergic condition
- Hot summers - April to October
- Children
- Intense itching
- " Morning Misery " - discomfort , discharge and blepharospasm on waking up

SPRING CATARRH/ VKC

- Cobblestone papillae
- Horner Trantas spots
- Shield ulcer







Active space

TREATMENT OF VKC

- Mast cell stabilizers – Na cromoglycate
- Topical steroids – Loteprednol, Prednisolone – may lead to cataracts and glaucoma

LAYERS OF THE CORNEA

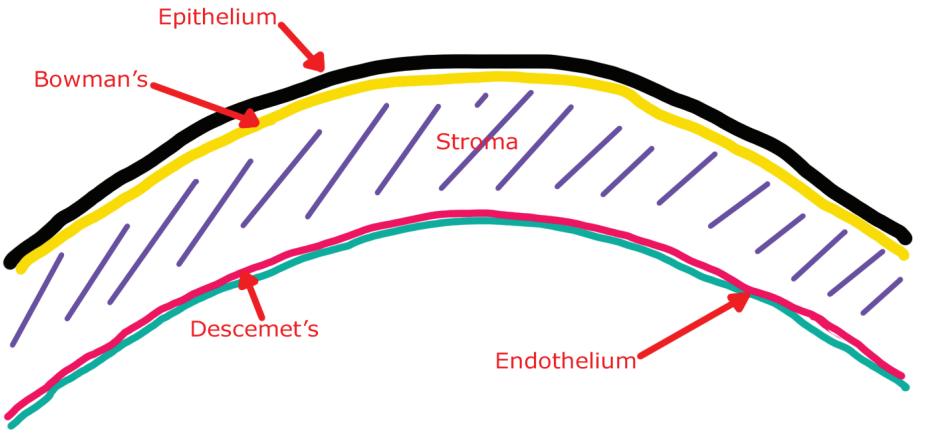
- Epithelium – protective barrier of the cornea
- Bowman's Membrane –cannot repair itself – scar formation
- Stroma –thickest layer of cornea – principial support
- Descemet's Membrane – most powerful membrane , blocks all microbes except –
- Endothelium – most important layer
- Maintains corneal transparency

Active space

LAYERS OF CORNEA

- Endothelium –
- Pumps aqueous out of cornea
- Na K ATPase endothelial pumps
- Endothelium irreparable – leads to corneal oedema
- PDL or Dua's layer- in between Descemet's and Stroma – thin , very strong membrane

LAYERS OF THE CORNEA

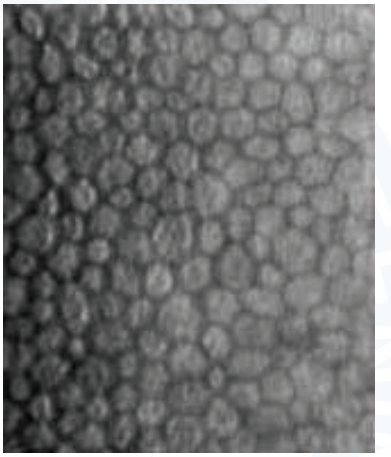


Active space

Corneal endothelium

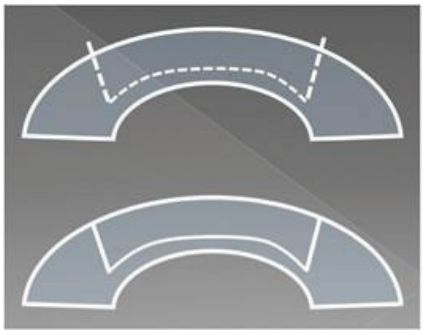
- Average count: 3000 cells/mm^2
- Every year 0.5% loss
- Critical density: <500 cells/mm^2

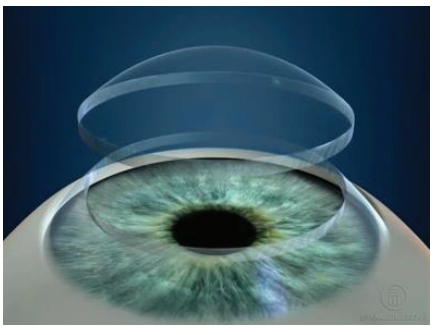
  For Corneal Donation >2000 cells/mm^2
- Counted by specular microscope



TYPES OF KERATOPLASTY

- Optical – for visual rehabilitation
- Penetrating keratoplasty PK – Full thickness
- Lamellar keratoplasty LK - Partial thickness

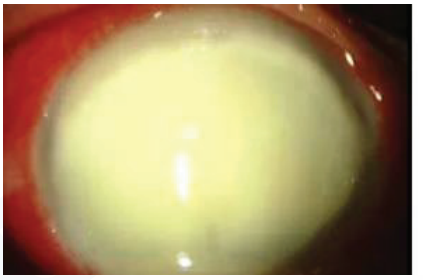




Active space

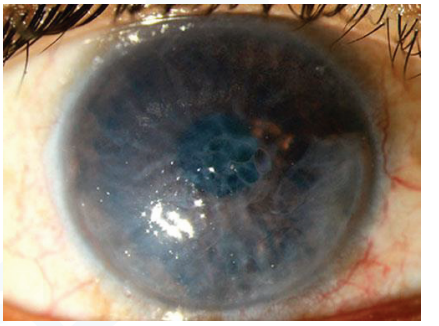
THERAPEUTIC KERATOPLASTY

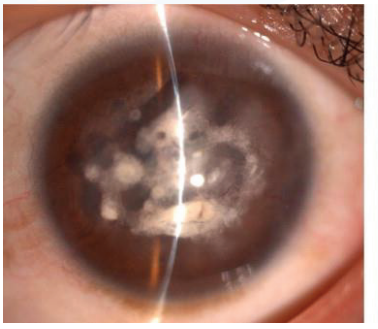
- Keratoplasty done to eliminate infection/ repair a structural defect

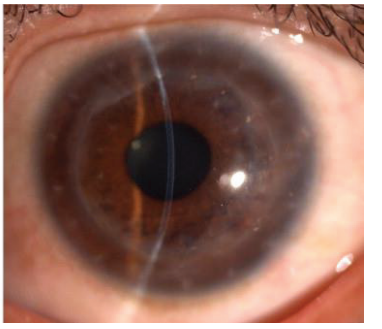


INDICATIONS OF KERATOPLASTY

- Pseudophakic bullous keratopathy
- Non healing corneal ulcers
- Corneal scars
- Keratoconus
- Corneal dystrophies







Active space

CORNEAL DONATION

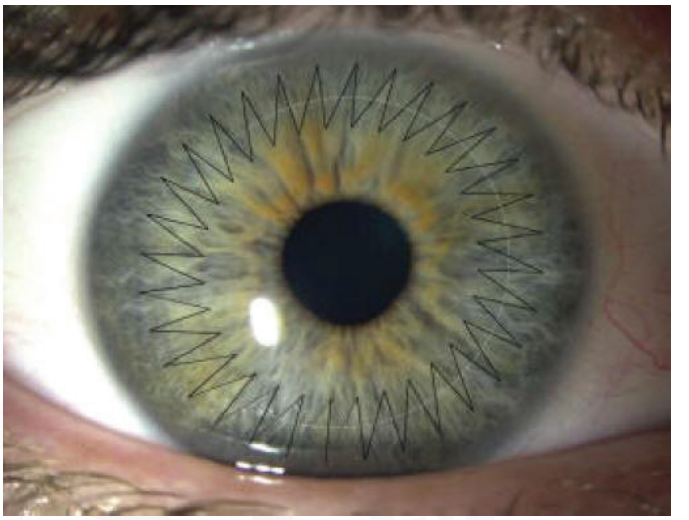
- HLA matching not required
- Within 6 hrs of death
- Preserved in MK (McCarey Kaufmann) media – 96 hrs
- Optisol GS - 14 days

CONTRAINDICATIONS

- HIV
- Hepatitis - B
- Rabies
- Bacterial / Fungal keratitis
- Herpes Simplex Virus
- Retinoblastoma
- Leukemias
- Death of unknown cause

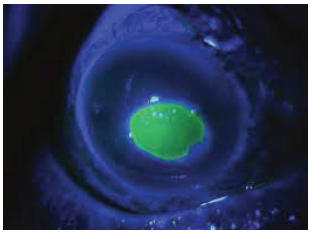
Active space

PK



CORNEAL ULCER

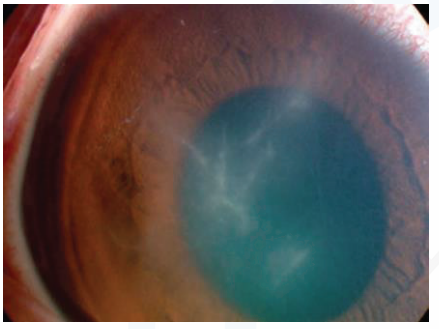
- Definition – a break in the epithelium with necrosis of underlying epithelium- vision threatening emergency
- Infectious – bacterial / viral /fungal / acanthamoeba
- Non infectious – neurotrophic / neuroparalytic / Mooren's ulcer / Vitamin A deficiency
- Symptoms – photophobia /pain /redness/ watering /loss of vision
- Differentiate between ulcerative and uveitis hypopyon



Active space

ACANTHAMOEBA KERATITIS

- Predisposing causes -Contact lens wearers
- Bathing in pools , hot tubs , swimming in contaminated water
- Ring shaped ulcer
- Pain out of proportion
- Radial keratoneuritis



BACTERIAL KERATITIS

Predisposing causes -Corneal trauma , CL wear , post refractive surgery

- M/c – Staph aureus /Pseudomonas
- M/C in India: Strep pneumoniae/ Pneumococcus
- Ulcus serpens

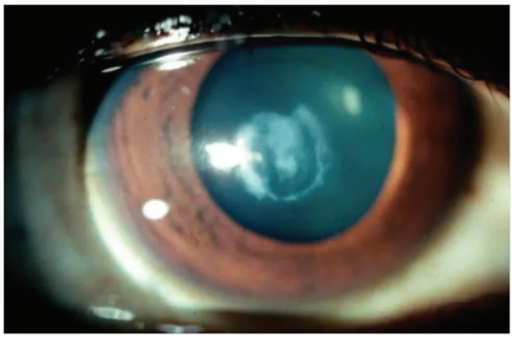
Active space

BACTERIA PENETRATING INTACT CORNEAL EPITHELIUM

- Corynebacterium diphtheriae
- Neisseria gonorrhea/meningitidis

FUNGAL KERATITIS

- M/c – Fusarium / Aspergillus
- Quiet ulcer , slow growing , asymptomatic
- Signs> symptoms



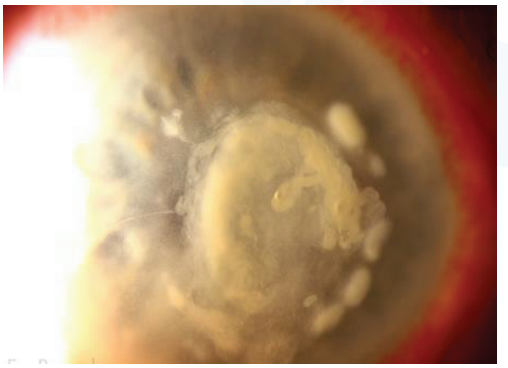
Active space

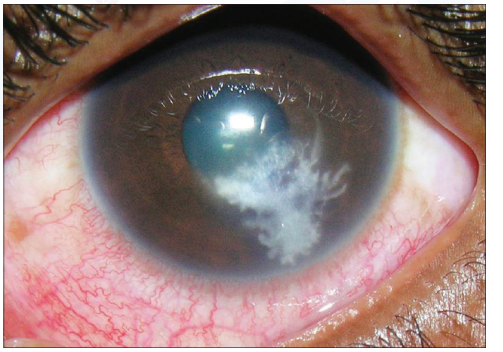
FUNGAL KERATITIS

Predisposing factors:

- Prolonged topical steroids
- Organic/vegetable matter injury

- Finger-like projections
- Feathery margins
- Satellite lesions

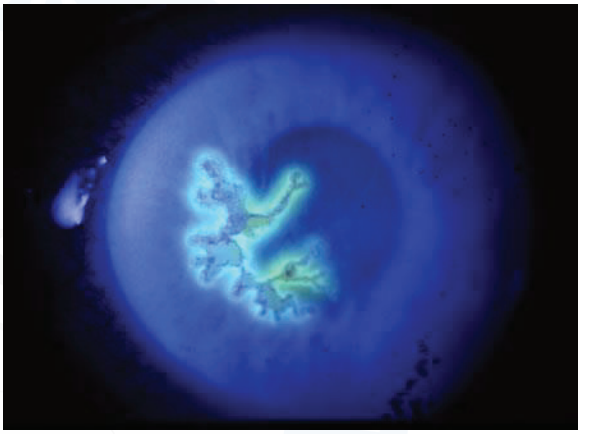




Active space

HSV KERATITIS - DENDRITIC ULCER

- Predisposing causes-reactivation of primary infection by stress , heat , trauma , CL
- Dendritic – branches of a tree
- Only true dendrites
- Loss of corneal sensations



HERPES ZOSTER OPHTHALMICUS-HZO SHINGLES

Reactivation of Varicella zoster(VZV) → chickenpox

Primary infection –Chickenpox , remains latent in dorsal root ganglia , reactivates later

- Old people >65, immuno compromised - DM, steroid therapy, anti cancer drugs, radiation
- Painful, itchy rash dermatome, NOT cross midline
- Young patient→ sign of HIV
- Hutchinson's sign – tip of nose affected → eye involved- supplied by same branch of nasociliary

Active space

MANAGEMENT

- Treatment to start within 72 hours
- M/c/c complication PHN – post herpetic neuralgia
- Famcyclovir 500 mg 3 times /day x 10 days
- Valacyclovir 1000 mg 3 times a day X 10 days

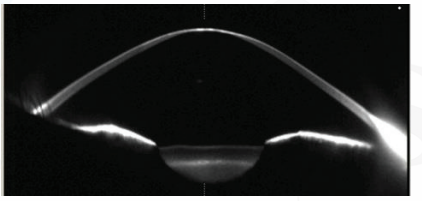
MANAGEMENT OF CORNEAL ULCERS

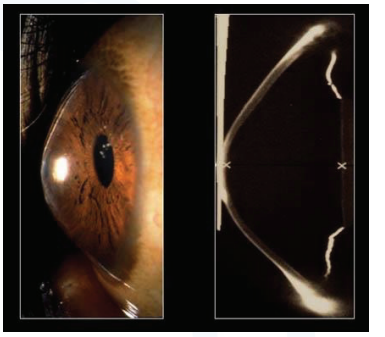
- Topical drugs only
- DOC of Bacterial ulcers - 4th Gen fluoroquinolones
- Moxifloxacin / Gatifloxacin
- DOC fungal ulcers – Natamicin
- DOC viral ulcers – Acyclovir
- DOC Acanthamoeba –PHMB / Propamidine
- Cycloplegics for pain – Atropine , homatropine

Active space

KERATOCONUS

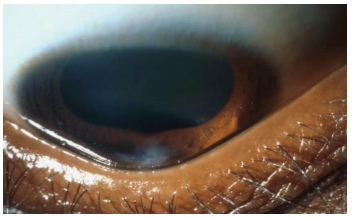
- Progressive thinning of central cornea → cone shaped bulge of cornea
- Risk factor → eye rubbing/ sleep apnoea
- Blurred vision → frequent change of glasses
- Diplopia
- High myopic astigmatism

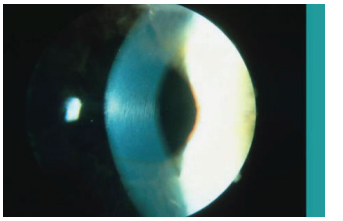




SIGNS AND SYMPTOMS

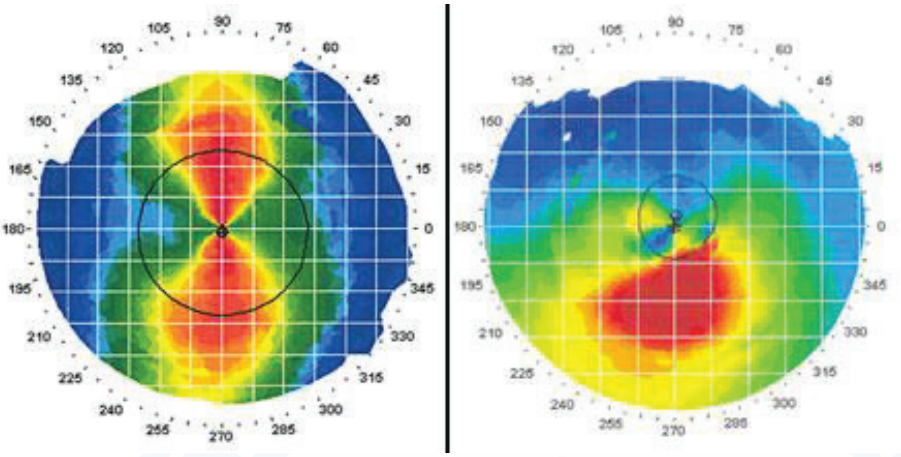
- Scissoring of retinoscopic reflexes
- Charleaux oil drop sign
- Munson's sign – protrusion of lower lid on downgaze
- Fleischer's ring – iron ring on base of cone
- Vogt's striae –vertical folds on stroma





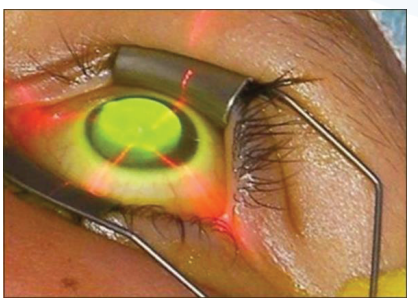
Active space

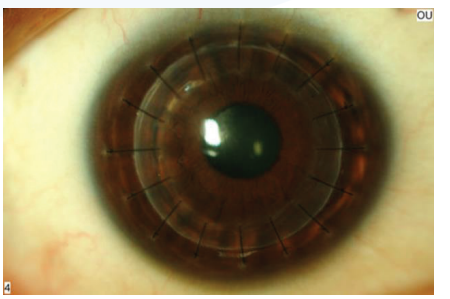
TOPOGRAPHY-INVESTIGATION OF CHOICE



TREATMENT

- Glasses
- Rigid gas permeable (RGP) contact lenses
- Keratoplasty- following stromal scarring after hydrops
- Corneal Collagen Crosslinking with Riboflavin(C3R)





Active space

Active space